

Construction of Higher Sheaf Theory

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Contents

1	Introduction	2
1.1	Motivation	2
1.2	A sketch of the thesis	3
1.3	Notation and convention	3
1.3.1	Category	3
1.3.2	Inclusions and embeddings	3
1.3.3	Higher algebra	4
2	$\text{Span}_n^{\text{enr}}(\mathcal{C}) \in n \text{ Cat}^{\text{Span}_1(\mathcal{C})}$	4
2.1	Span_n	4
2.2	Symmetric monoidal structure on Span	8
2.3	$B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons (\text{Cat}_n)_* / : \Omega$	12
2.4	$\text{LMod}_{B A}(\text{Cat}^{\mathcal{M}}) \cong \text{Cat}^{\text{LMod}_A(\mathcal{M})}$	16
2.5	$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \hookrightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$	19
2.6	$\text{Span}_n^{\text{enr}}(\mathcal{C}) \in n \text{ Cat}^{\text{Span}_1(\mathcal{C})}$	22
3	$\text{Span}_n^{\text{enr}}(\mathcal{C}) \rightarrow (n-1) \text{ Cat}^{\text{Span}_1(\mathcal{C})}$	28
3.1	$\Gamma : \mathcal{P}_{(n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \rightarrow (n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$	28
3.2	A lax symmetric monoidal (∞, n) functor	29
4	$2 \text{ Shv}(\text{BG})$	32
4.1	$2 \text{ Shv}(-)$	32
4.2	$2 \text{ Shv}(\text{BG})$	34
5	Appendix on enriched category theory	36
5.1	Enriched category theory	36
5.2	Fibrations	40
5.3	Symmetric monoidal structure	42
5.4	Loop and deloop	46

1 Introduction

1.1 Motivation

Braided monoidal category structure appears as a fundamental framework to understand knot theory and topological field theory; it can be viewed as a special case of $\mathbb{E}_2$ algebra object in the $(2, 1)$ category $\text{Cat}_{\text{ordinary}}$ of ordinary categories. By Dunn additivity, we can characterize the $(\infty, 1)$ operad $\mathbb{E}_2$ as the Boardman-Vogt tensor product $\mathbb{E}_1 \otimes \mathbb{E}_1$ of $\mathbb{E}_1$ operad.

There is an ongoing project on $(\infty, 2)$ operad theory, developed by Tallak Manum, and by taking the Gray-Boardman-Vogt tensor product of $\mathbb{E}_1$ with itself, we get an $(\infty, 2)$ operad $\mathbb{E}_2^{\text{lax}}$. Moreover, one can characterize the $\mathbb{E}_2^{\text{lax}}$ algebra in the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$ in a nice way, by the work of Quoc Ho, Walker H. Stern, and Jonte Gödicke. The space of $\mathbb{E}_2^{\text{lax}}$ objects in $\text{Span}_2(\mathcal{C})$ is precisely the space of 2-Segal objects in $\mathcal{C}$, which is a ‘lax’ and ‘ $\mathbb{E}_2$ ’ generalization of the results of the $\mathbb{E}_1$ algebra object in $\text{Span}_1(\mathcal{C})$ in the PHD thesis of Jonte Gödicke.

To transfer the $\mathbb{E}_2^{\text{lax}}$ algebra objects, we need the ‘lax symmetric monoidal $(\infty, 2)$ functor’, which is a generalization of the lax symmetric monoidal $(\infty, 1)$ functor developed in [25] by Lurie. The symmetric monoidal $(\infty, 2)$ functor can be defined as the morphism in the category $\text{CAlg}(\text{Cat}_2)$ where Cat_2 is equipped with the Cartesian symmetric monoidal structure. By the (Un)straightening of $(\infty, 2)$ categories, any symmetric monoidal $(\infty, 2)$ category $\mathcal{C}$ unstraightens to an $(\infty, 2)$ coCartesian fibration $\mathcal{C}^\otimes$ over $\text{Comm}^\otimes$, and the symmetric monoidal functor is a morphism between the associated $(\infty, 2)$ coCartesian fibrations.

Definition 1.1.1. A 2 functor $\pi : C \rightarrow D$ is a 2-coCartesian fibration if it satisfies the condition:

- (coCartesian 1 morphism) Given any $c_1 \in C$, and 1 morphism $e : \pi(c_1) \rightarrow d$ in D , there exists a 1 morphism $f : c_1 \rightarrow c_2$ such that $\pi(f) = e$ and f induces a pullback

$$\begin{array}{ccc} \text{Map}_C(c_2, x) & \longrightarrow & \text{Map}_C(c_1, x) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}_D(\pi(c_2), \pi(x)) & \longrightarrow & \text{Map}_D(\pi(c_1), \pi(x)) \end{array} \quad \text{for any } x \in C.$$

diagram:

- (Cartesian 2 morphism) $\text{Map}_C(x, y) \rightarrow \text{Map}_D(\pi(x), \pi(y))$ is a Cartesian fibration.
- The composition map $\text{Map}_C(x, y) \times \text{Map}_C(y, z) \rightarrow \text{Map}_C(x, z)$ preserves Cartesian edges.

And we can define the lax symmetric monoidal $(\infty, 2)$ functor as

Definition 1.1.2. A lax symmetric monoidal $(\infty, 2)$ functor $G : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ is a functor that preserves the inert coCartesian 1 morphisms lifting and Cartesian 2 morphisms.

By [2], we have:

Theorem 1.1.3. Given an adjunction $f : \mathcal{C} \rightleftharpoons \mathcal{D} : g$ in Cat_2 , if the left adjoint f can be enhanced into a symmetric monoidal $F : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ in $\text{CAlg}(\text{Cat}_2)$, then the right adjoint can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor $G : \mathcal{D}^\otimes \rightarrow \mathcal{C}^\otimes$.

On the other hand, a sheaf theory can be encoded into a lax symmetric monoidal functor $\text{Shv}(-) :$

$\text{Span}_1(\mathcal{C}) \rightarrow \text{Pr}^L$, in the recent work on higher sheaf theory by Peter Scholze and H rman Stefanich, for stacky objects, the higher sheaf theory recovers more geometric information than 1 sheaf theory. In this thesis, we construct a lax symmetric monoidal $(\infty, 2)$ functor $2\text{Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_\kappa$ for some fixed regular cardinality κ , assigning the higher sheaf. And we can construct interesting $\mathbb{E}_2$ and $\mathbb{E}_2^{\text{lax}}$ algebra objects in 2Pr_κ through the functor $2\text{Shv}(-)$.

1.2 A sketch of the thesis

In section 2, we supplement the details of [37] by Stefanich, showing that $\text{Span}_n(\mathcal{C}) \in \text{Cat}_n$ can be enhanced into an object in $n\text{Cat}^{\text{Span}_1(\mathcal{C})}$, denoted as $\text{Span}_n^{\text{enr}}(\mathcal{C})$. In section 3, we show that there exists a lax symmetric monoidal (∞, n) functor from $\text{Span}_n^{\text{enr}}(\mathcal{C}) \rightarrow (n-1)\text{Cat}^{\text{Span}_1(\mathcal{C})}$. As an application, in section 4, we develop an $(\infty, 2)$ six functor formalism $2\text{Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}$ from a given six functor formalism $\text{Shv}(-) : \text{Span}_1(\mathcal{C}) \rightarrow 1\text{Pr}$, and discuss its value on the classifying stack BG of a smooth algebraic group scheme. And in the appendix, there are some necessary background materials on enriched ∞ -category theory.

1.3 Notation and convention

1.3.1 Category

- Let Cat_n denote the ∞ -category of (∞, n) categories. When $n = 1$, let Cat denote Cat_1 .
- Let Cat^{pb} denote the ∞ -categories with fiber products and ∞ -functors preserving fiber products.
- Let Spc denote the ∞ -category of Kan complexes.
- Given an ∞ -category $\mathcal{C}$, let $\mathcal{C}^\simeq$ denote the maximal Kan complex of $\mathcal{C}$.
- Let Pr^L denote the ∞ -category of presentably ∞ -categories as defined in [24, definition 4.4.0.1].
- Let 1Pr denote the ∞ -category of $\aleph_1$ -compactly generated categories as defined in [24, definition 5.5.7.1], which itself is a presentably category.

1.3.2 Inclusions and embeddings

- We often call a fully faithful functor $\mathcal{C} \rightarrow \mathcal{D}$ an embedding.
- We call a functor $\phi : \mathcal{D} \rightarrow \mathcal{D}$ an inclusion or say that ϕ exhibits $\mathcal{C}$ as a subcategory of $\mathcal{D}$ if for any ∞ -category E the induced map $\text{Cat}_\infty(E, \mathcal{C}) \rightarrow \text{Cat}_\infty(E, \mathcal{D})$ is an embedding. A functor $\phi : \mathcal{C} \rightarrow \mathcal{D}$ is an inclusion if and only if it induces an embedding on maximal subspaces and on all mapping spaces.
- We say that an inclusion $\phi : \mathcal{C} \rightarrow \mathcal{D}$ exhibits $\mathcal{C}$ as a wide subcategory of $\mathcal{D}$ if ϕ is essentially surjective.

1.3.3 Higher algebra

- Let Op be the category of operads as defined in [25, chapter 2].
- Let $\text{Ass} \cong \Delta^{op}$ be the approximation of the operad $\text{Assoc}^{\otimes}$ in [25, definition 4.1.1.3].
- Let Op^{ns} be the category of non-symmetric operads, as defined in [15, notation 2.36], which is equivalent to the slice category of Op over $\text{Assoc}^{\otimes}$ by [25, theorem 4.3.14].
- Let Mon^{lax} be the full category of Op^{ns} consisting of monoidal categories.
- Let Mon be the wide subcategory of Mon^{lax} with 1-morphisms being monoidal functors.
- Given two operads $\mathcal{V}$ and $\mathcal{W}$, let $\text{Alg}_{\mathcal{V}}(\mathcal{W})$ denote the ∞ -category of operad maps between $\mathcal{V}$ and $\mathcal{W}$. When $\mathcal{V} = \text{Comm}^{\otimes}$, let $\text{CAlg}(\mathcal{W})$ denote $\text{Alg}_{\mathcal{V}}(\mathcal{W})$.
- We call an object $\mathcal{C} \in \text{CAlg}(\text{Cat}_n)$ a symmetric monoidal (∞, n) category.
- We call a morphism $F : \mathcal{C} \rightarrow \mathcal{D}$ in $\text{CAlg}(\text{Cat}_n)$ a symmetric monoidal (∞, n) functor. Especially, the right adjoint of a symmetric monoidal (∞, n) functor is a lax symmetric monoidal (∞, n) functor.

2 $\text{Span}_n^{enr}(\mathcal{C}) \in n \text{Cat}^{\text{Span}_1(\mathcal{C})}$

2.1 Span_n

In this subsection, we review the constructions of $\text{Span}_n(\mathcal{C})$ and $\text{Span}_{n-1}^+(\mathcal{C})$ in [12, section 5].

Construction 2.1.1. Consider the following functor

$$\overline{\text{SPAN}}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n} \times \text{id}} (\text{Cat}^{op})^{\times n} \times \text{Cat}^{pb} \xrightarrow{(\times^n) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \xrightarrow{(-)^{\simeq}} \text{Spc}$$

On the other hand, noticing the natural projection $\text{Tw}(-) \Rightarrow \text{id}$, we get a subfunctor of $\overline{\text{SPAN}}_n$

$$\overline{\text{SPAN}}_{(n-1)}^+ : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{(\text{Tw}(-)^{\times n-1} \times \text{id}) \times \text{id}} (\text{Cat}^{op})^{\times n} \times \text{Cat}^{pb} \rightarrow \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \rightarrow \text{Spc}$$

Moreover, denote $\Lambda_n \subset \text{Tw}(n)$ as the full subcategory consisting of $(i \leq j)$ with $j - i \leq 1$, by taking the subspace

$$\text{Map}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n), \mathcal{C})^{\text{cart}} \subset \text{Fun}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n), \mathcal{C})^{\simeq}$$

consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n) \rightarrow \mathcal{C}$ which are the right Kan extension of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_n}}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_n} \rightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n)$, we get a subfunctor

$$\text{SPAN}_n \hookrightarrow \overline{\text{SPAN}}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$$

Given $\mathcal{C} \in \text{Cat}^{pb}$, there is an associated multisimplicial space $\text{SPAN}_n(\mathcal{C}) : (\Delta^{op})^n \rightarrow \text{Spc}$, and by [12, proposition 5.14] and [8, proposition 4.9], there is:

Lemma 2.1.2. For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_n(\mathcal{C})$ is an n -fold category object in Spc .

On the other hand, for $\overline{\text{SPAN}}_{n-1}^+$, we acquire a subfunctor:

$$\text{SPAN}_{n-1}^+ \hookrightarrow \overline{\text{SPAN}}_{n-1}^+ : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$$

by taking the subspace

$$\text{Map}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n], \mathcal{C})^{cart} \subset \text{Map}([m_n], \text{Fun}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}), \mathcal{C}))$$

consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n] \rightarrow \mathcal{C}$ that are the right Kan extension of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \times [m_n]}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \times [m_n] \rightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n]$.

And especially by [12, lemma 5.6], there is a subfunctor inclusion $\text{SPAN}_{n-1}^+ \rightarrow \text{SPAN}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$. For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_+(\mathcal{C}) : (\Delta^{op})^n \rightarrow \text{Spc}$ can be regarded as a functor $(\Delta^{op})^{n-1} \rightarrow \text{Fun}(\Delta^{op}, \text{Spc})$. There is

Lemma 2.1.3. For $\mathcal{C} \in \text{Cat}^{pb}$,

- (1) $\overline{\text{SPAN}}_{n-1}^+(-) : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ corresponds to a functor $\overline{\text{SPAN}}_{n-1}^+(-)' : (\Delta^{op})^{n-1} \times \text{Cat}^{pb} \rightarrow \text{Cat} \subset \text{Fun}(\Delta^{op}, \text{Spc})$ under the Hom-tensor adjunction, which coincides with the functor

$$(\Delta^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id}} (\text{Cat}^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\times^{n-1} \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat}$$

- (2) And $\text{SPAN}_{n-1}^+(-) : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ corresponds to the subfunctor of $\overline{\text{SPAN}}_{n-1}^+(-)'$ consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \rightarrow \mathcal{C}$ which are the right Kan extensions of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}}}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \hookrightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1})$

- (3) For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_{n-1}^+(\mathcal{C})$ is a n -fold category object in Spc and $\text{SPAN}_{n-1}^+(\mathcal{C})([m_1], \dots, [m_{n-1}], -) : \Delta^{op} \rightarrow \text{Spc}$ is a complete Segal space for any $([m_1], \dots, [m_{n-1}]) \in (\Delta^{op})^{n-1}$.

- (4) Moreover, there is a commutative diagram

$$\begin{array}{ccc} (\Delta^{op})^{n-1} \times \text{Cat}^{pb} & \xrightarrow{\text{SPAN}_{n-1}^+(-)'} & \text{Cat} \\ & \searrow \text{SPAN}_{n-1} & \swarrow (-)^{\simeq} \\ & \text{Spc} & \end{array}$$

Proof. (1) and (2) are due to the following facts:

- Noticing the following commutative diagram

$$\begin{array}{ccc} \text{Cat}^{op} \times \text{Cat}^{op} \times \text{Cat} & \xrightarrow{\text{id} \times \text{Fun}(-, -)} & \text{Cat}^{op} \times \text{Cat} \\ \downarrow (\times) \times \text{id} & & \downarrow \text{Fun}(-, -) \\ \text{Cat}^{op} \times \text{Cat} & \xrightarrow{\text{Fun}(-, -)} & \text{Cat} \end{array}$$

- Noticing that the functor

$$\text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

corresponds to the functor

$$\Delta^{op} \times \text{Cat} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \xrightarrow{(-)^\approx} \text{Spc}$$

- Thus, the functor

$$(\Delta^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id}} (\text{Cat}^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{(-)^{n-1} \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

corresponds exactly to the functor

$$\overline{\text{SPAN}}_{n-1}^+ : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id} \times \text{id}} (\text{Cat}^{op})^{\times n-1} \times \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{(\times) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \rightarrow \text{Spc}$$

under the Hom-tensor adjunction.

(3) and (4) are direct consequences of (1) and (2). □

By [12, proposition 4.12], there is a right adjoint functor

$$\mathcal{U}_{\text{Seg}_n} : \text{Cat}_n(\text{Spc}) \rightarrow \text{Seg}_n(\text{Spc})$$

from the category of n -fold category objects in Spc to the category of n -Segal spaces, furthermore, there is a left adjoint functor

$$L : \text{Seg}_n(\text{Spc}) \rightarrow \text{CSS}_n(\text{Spc}) \cong \text{Cat}_n$$

from the category of n -Segal space to the category of complete n -Segal space, which is one model of (∞, n) categories.

What's more, one can show that after taking $\mathcal{U}_{\text{Seg}_n}$, for $\text{SPAN}_n(\mathcal{C})$ and $\text{SPAN}_{n-1}^+(\mathcal{C})$, they have already been the complete n -Segal space:

Lemma 2.1.4. *For $\mathcal{C} \in \text{Cat}^{pb}$, and $n \geq 1$, both $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_n(\mathcal{C}))$ and $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_{n-1}^+(\mathcal{C}))$ are complete n -Segal spaces.*

Proof. The case of $\mathcal{U}_{\text{Seg}}(\text{SPAN}_n(\mathcal{C}))$ is by [12, corollary 8.5].

As for $\text{SPAN}_{n-1}(\mathcal{C})$, we follow the same proof strategy:

- By induction, the case when $n = 1$ is trivial.
- And for $n \geq 2$, by the same proof of [12, proposition 8.3], for any $x, y \in \mathcal{C}$ $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})(x, y) \cong \mathcal{U}_{\text{Seg}_{n-1}} \text{SPAN}_{n-2}^+(\mathcal{C}_{/x \times y})$, which is complete by induction.
- Thus, both $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})(x, y) \cong \mathcal{U}_{\text{Seg}_{n-1}} \text{SPAN}_{n-2}^+(\mathcal{C}_{/x \times y})$ and $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})_{\bullet, 0, \dots, 0} \cong \mathcal{U}_{\text{Seg}_1} \text{SPAN}_1(\mathcal{C})$ are complete, and $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_{n-1}^+(\mathcal{C}))$ is also complete by [12, theorem 7.18]

□

Remark 2.1.5. Especially, the functor $\mathcal{U}_{\text{Seg}_n}$ is just taking sub-multisimplicial space, thus all the inclusion relations for SPAN_{n-1}^+ and SPAN_n still hold.

By composition with $\mathcal{U}_{\text{Seg}_n}$, there are functors

$$\begin{aligned}\text{Span}_n(-) &= \mathcal{U}_{\text{Seg}_n} \circ \text{SPAN}_n : \text{Cat}^{pb} \rightarrow \text{Cat}_n \\ \text{Span}_{n-1}^+(-) &= \mathcal{U}_{\text{Seg}_n} \circ \text{SPAN}_{n-1}^+ : \text{Cat}^{pb} \rightarrow \text{Cat}_n\end{aligned}$$

and there is a natural inclusion $\iota_{n-1} : \text{Span}_{n-1}^+(-) \Rightarrow \text{Span}_n(-)$.

Remark 2.1.6. By the above lemmas, there is a commutative diagram:

$$\begin{array}{ccc} \text{Cat}^{pb} & \xrightarrow{\text{Span}_{n-1}^+(-)} & \text{Cat}_n \\ & \searrow \text{Span}_{n-1}(-) \quad \swarrow \tau_n & \\ & \text{Cat}_{n-1} & \end{array}$$

, by taking the right adjoint of the inclusion-truncation adjunction

$$\tau_n : \text{Cat}_n \rightleftarrows \text{Cat}_{n-1} : i$$

, there is a natural transformation $\iota_+ : \text{Span}_{n-1}(-) \Rightarrow \text{Span}_{n-1}^+(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n$, which value at $\mathcal{C} \in \text{Cat}^{pb}$ being the inclusion $\text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$, thus ι_{n-1}^+ exhibits $\text{Span}_{n-1}(-)$ as a subfunctor of $\text{Span}_{n-1}^+(-)$.

Corollary 2.1.7. *In conclusion, there are inclusions of functors:*

$$\text{Span}_{n-1}(-) \xrightarrow{\iota_{n-1}^+} \text{Span}_{n-1}^+(-) \xrightarrow{\iota_{n-1}} \text{Span}_n^+(-)$$

.

There are some identifications we can make in comparison with the construction in [18] and [35]:

Remark 2.1.8. The functor $\text{Span}_1(-) : \text{Cat}^{pb} \rightarrow \text{Cat}$ coincides with the functor $\text{Corr}(-) : \text{GeomSetup} \rightarrow \text{Cat}$ by the embedding $\text{Cat}^{pb} \hookrightarrow \text{GeomSetup}$.

Remark 2.1.9. Noticing that $\overline{\text{SPAN}}_1^+ : \Delta^{op} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ coincides with the functor in [35, construction 3.2.1]:

$$\overline{2\text{Corr}}(-) : \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

, thus the functor $2\text{Corr}(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_2$ loc. cit is exactly the functor $\text{Span}_1^+(-)$.

2.2 Symmetric monoidal structure on Span

In this subsection, we use the functoriality of $\text{Span}_n(-)$ to enhance $\text{Span}_n(\mathcal{C})$ into a symmetric monoidal (∞, n) category.

Construction 2.2.1. By [7, lemma 4.1 & remark 4.1], the functors

$$\text{Span}_n(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n, \text{Span}_{n-1}^+(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n$$

preserve finite limits; thus, they are enhanced into symmetric monoidal functors

$$\text{Span}_n(-) : (\text{Cat}^{pb})^\times \rightarrow (\text{Cat}_n)^\times, \text{Span}_{n-1}^+ : (\text{Cat}^{pb})^\times \rightarrow (\text{Cat}_n)^\times$$

And for $\mathcal{C} \in \text{Cat}^{pb}$, consider the Cartesian symmetric monoidal structure on $\mathcal{C}$. We get $\mathcal{C} \in \text{CAlg}(\text{Cat}^{pb}) \mapsto \text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$; thus, $\text{Span}_n(\mathcal{C})$ is enhanced into a symmetric monoidal (∞, n) category; so is $\text{Span}_{n-1}^+(\mathcal{C})$.

Moreover, consider the subfunctor $\iota : \text{Span}_1^+(-) \Rightarrow \text{Span}_2(-)$, which is encoded as a functor $\iota : [1] \rightarrow \text{Fun}(\text{Cat}^{pb}, \text{Cat}_2)$ that factors through the full subcategory $\text{Fun}^\times(\text{Cat}^{pb}, \text{Cat}_2) \subset \text{Fun}(\text{Cat}^{pb}, \text{Cat}_2)$ consisting of the finite-product-preserving functors. By [25, proposition 2.4.1.6], $\iota : [1] \rightarrow \text{Fun}^\times(\text{Cat}^{pb}, \text{Cat}_2) \cong \text{Fun}^\otimes((\text{Cat}^{pb})^\times, (\text{Cat}_2)^\times)$, thus $\iota : \text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})$ can be enhanced into a symmetric monoidal (∞, n) functor.

$$\iota_{n-1} : \text{Span}_{n-1}^+(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})$$

Likewise, $\iota_{n-1}^+ : \text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$ can also be enhanced into a symmetric monoidal (∞, n) functor.

Remark 2.2.2. A symmetric monoidal (∞, n) category $\mathcal{M}$ is equivalent to a functor $\text{Comm}^\otimes \rightarrow \text{Cat}_n$. By the (∞, n) (un)straightening (for example, constructed in [29]), it corresponds to an (∞, n) coCartesian fibration $\mathcal{M}^\otimes \rightarrow \text{Comm}^\otimes$. Under this correspondence,

$$\iota_{n-1} : \text{Span}_{n-1}^+(\mathcal{C})^\otimes \rightarrow \text{Span}_n(\mathcal{C})^\otimes$$

can be characterized as the non-full (∞, n) subcategory inclusion, where it contains all the coCartesian liftings, and inside a fixed fiber, it contains the n -morphisms in the form of a span, with the left leg being an equivalence. Likewise,

$$\iota_{n-1}^+ : \text{Span}_{n-1}(\mathcal{C})^\otimes \rightarrow \text{Span}_{n-1}^+(\mathcal{C})^\otimes$$

can be characterized as the non-full (∞, n) subcategory inclusion, where it contains all the coCartesian liftings, and inside a fixed fiber, it contains only the invertible n -morphisms. In conclusion, we can regard ι_{n-1} and ι_{n-1}^+ as symmetric monoidal (∞, n) functors which underlying (∞, n) functors are non-full inclusions.

Remark 2.2.3. By [remark 2.1.9](#), we can see that the symmetric monoidal structure on $\text{Span}_1^+(\mathcal{C})$ coincides with the symmetric monoidal structure on $2\text{Corr}(\mathcal{C})$ in [35, remark 3.2.3].

Remark 2.2.4. Moreover, the composition of symmetric monoidal inclusions $\mathcal{C} \xrightarrow{\iota_0} \text{Span}_1(\mathcal{C}) \xrightarrow{\iota_1^+} \text{Span}_1^+(\mathcal{C})$ is equivalent to the symmetric monoidal inclusion $\iota_{\mathcal{C}} : \mathcal{C} \rightarrow 2\text{Corr}(\mathcal{C})$ constructed in [35, Construction 3.2.4]. Both inclusions are characterized as the same non-full $(\infty, 1)$ subcategory of the $(\infty, 2)$ category $\text{Span}_1^+(\mathcal{C})^{\otimes}$ over $\text{Comm}^{\otimes}$, thus satisfying the universal property established in [35, corollary 1.2.2]: a symmetric monoidal $(\infty, 2)$ functor $\mathcal{C} \rightarrow \mathcal{D}$ admits a unique factorization through $\text{Span}_1^+(\mathcal{C})$ if it satisfies the Beck-Chevalley condition .

On the other hand, in [18, section 2.3], there is a symmetric monoidal structure constructed on a geometric setup (C, C_0) :

$$\text{Span}_1(C, C_0)^{\otimes} := \text{Span}_1(C^{op, \Pi})^{op}, (C_0^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \rightarrow \text{Comm}^{\otimes}$$

which is an ∞ -operad and is a symmetric monoidal ∞ -category when C admits finite products. And by [18, example 2.3.6], when a morphism between geometric setups $(f, f_0) : (C, C_0) \rightarrow (D, D_0)$ satisfies that f preserves finite products, then the induced morphism $\text{Span}_1(C, C_0)^{\otimes} \rightarrow \text{Span}_1(D, D_0)^{\otimes}$ is symmetric monoidal.

Especially, by taking the geometric setup to be $(\mathcal{C}, \mathcal{C})$, we get another symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$, and in the remaining part, we are going to show that this is equivalent to the one acquired from the functoriality of $\text{Span}_1(-)$.

Remark 2.2.5. • In [18, proposition 2.3.5], the functor $\text{Span}_1(-)^{\otimes} : \text{GeomSetup} \rightarrow \text{Op}$ is constructed in the following way:

$$\begin{array}{ccc} \text{GeomSetup} & \xrightarrow{C \mapsto ((C^{op, \Pi})^{op})} & \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} \xrightarrow{\text{Span}_1(-)} \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} \\ & & \text{pullback along } \text{Comm}^{\otimes} \rightarrow \text{Span}_1(\text{Comm}^{\otimes, op}) \xrightarrow{\quad} \text{Cat}_{/\text{Comm}^{\otimes}} \end{array}$$

- Notice that when $C \rightarrow D$ preserves pullback, given $e : d \rightarrow c \in C$, there is a commutative diagram:

$$\begin{array}{ccc} C/c & \longrightarrow & D/F(c) \\ e^* \downarrow & & F(e)^* \downarrow \\ C/d & \longrightarrow & D/F(d) \end{array}$$

- And we apply the above observation to the pullback-preserving functor $\text{Span}_1(-)$, and $e : (\langle 1 \rangle, \langle 1 \rangle) \rightarrow (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})$, then there is a commutative diagram:

$$\begin{array}{ccccc} \text{GeomSetup} & & \text{Span}_1(-)^{\otimes} & & \\ \downarrow (-)^{op, \Pi, op} & & \searrow & & \\ \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} & \xrightarrow{\quad} & \text{Op} \subset \text{Cat}_{/\text{Comm}^{\otimes}} \\ \downarrow e^* & & \downarrow \text{Span}_1(e)^* & & \swarrow \\ \text{GeomSetup}_{/*=(\langle 1 \rangle, \langle 1 \rangle)} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/*=\{\langle 1 \rangle\}} & & \end{array}$$

- Thus we have a commutative diagram:

$$\begin{array}{ccccc}
 \text{Cat}^{pb} & \xrightarrow{\text{Span}(-)^{\otimes}} & \text{Cat}^{\otimes} & \hookrightarrow & \text{Op} \\
 & \searrow \text{Span}(-) & \searrow & & \downarrow G \\
 & & & & \text{Cat}
 \end{array}$$

where:

- G is induced by pullback along $* \cong \{1\} \rightarrow \text{Comm}^{\otimes}$ and sends an ∞ -operad to its underlying category.
- Moreover, after precomposition with $\text{Cat}^{pb} \rightarrow \text{GeomSetup}$, the functor assigning the symmetric monoidal structure $\text{Span}_1(-)^{\otimes}$:

$$\text{Span}(-)^{\otimes} : \text{GeomSetup} \rightarrow \text{Op}$$

factors through the non full subcategory of Op consisting of symmetric monoidal ∞ -categories and symmetric monoidal ∞ -functors between them.

- Thus, there is a factorization

$$\text{Span}_1(-)^{\otimes} : \text{Cat}^{pb} \rightarrow \text{Cat}^{\otimes} \subset \text{Op}$$

.

Moreover, there is:

Lemma 2.2.6. *The above functor $\text{Cat}^{pb} \rightarrow \text{GeomSetup} \xrightarrow{\text{Span}_1(-)^{\otimes}} \text{Cat}^{\otimes}$, $C \mapsto (C, C) \mapsto \text{Span}_1(C)^{\otimes}$ preserves finite products.*

Proof. • Obviously $\text{Span}_1(*)^{\otimes} \cong \text{Span}_1(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \cong \text{Comm}^{\otimes}$, thus $\text{Span}_1(-)^{\otimes}$ preserves the final object. And it remains to show that it preserves products.

- Recall the construction of the coCartesian symmetric monoidal structure in [25, Construction 2.4.3.1]. Given an ∞ category C , C^{II} is defined as

$$\text{Hom}_{\text{N}(\text{Fin})_*}(K, C^{\text{II}}) \cong \text{Hom}_{\text{sSet}}(K \times_{\text{N}(\text{Fin})_*} \Gamma^*, C)$$

. it's obvious that for $C \times D$, the representator on the left hand side should be $C^{\text{II}} \times D^{\text{II}}$, which implies $(C \times D)^{\text{II}} \cong (C^{\text{II}} \times_{\text{Comm}^{\otimes}} D^{\text{II}})$, where the second product takes place in $\text{Cat}_{/\text{N}(\text{Fin})_*}$. Thus, it's the fiber product over $\text{N}(\text{Fin})_*$.

- So there are equivalences in Cat :

$$((C \times D)^{op, \text{II}})^{op} \cong (C^{op, \text{II}} \times_{\text{Comm}^{\otimes}} D^{op, \text{II}})^{op} \cong (C^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \text{II}})^{op}$$

, and so are

$$((C \times D)^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong ((C^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \text{II}})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}$$

and

$$\begin{aligned} & ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \Pi})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \\ & ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \times_{\text{Comm}^{\otimes, op}} ((D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \end{aligned}$$

by the fact that $\text{Comm}^{\otimes, \simeq} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \text{Comm}^{\otimes, \simeq}$ and limit commutes with limit.

- Thus we have a fiber product in GeomSetup :

$$\begin{array}{ccc} (((C \times D)^{op, \Pi})^{op}, ((C \times D)^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \longrightarrow & ((C^{op, \Pi})^{op}, (C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \\ \downarrow \lrcorner & & \downarrow \\ ((D^{op, \Pi})^{op}, (D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \longrightarrow & (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \end{array}$$

•

- Since $\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$ preserves finite limits, thus there is a pullback diagram in Cat :

$$\begin{array}{ccc} \text{Span}_1(C \times D)^{\otimes} & \longrightarrow & \text{Span}_1(C)^{\otimes} \\ \downarrow \lrcorner & & \downarrow \\ \text{Span}_1(D)^{\otimes} & \longrightarrow & \text{Comm}^{\otimes} \end{array}$$

here we identify $\text{Span}_1(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \cong \text{Comm}^{\otimes}$ by [18, example 2.2.13].

- This implies we have a product in $\text{Cat}^{\otimes} \subset \text{Op}$:

$$\text{Span}_1(C \times D)^{\otimes} \cong \text{Span}_1(C)^{\otimes} \times_{\text{Comm}^{\otimes}} \text{Span}_1(D)^{\otimes}$$

- Thus $\text{Span}_1(-)^{\otimes} : \text{Cat}^{pb} \rightarrow \text{Cat}^{\otimes}$ preserves products.

□

Thus $\text{Cat}^{pb} \xrightarrow{\text{Span}(-)^{\otimes}} \text{Cat}^{\otimes}$ preserves products, and so is the functor G by [25, remark 2.1.4.10], thus there is a commutative diagram:

$$\begin{array}{ccc} \text{Cat}^{pb, \times} & \xrightarrow{\text{Span}(-)^{\otimes}} & (\text{Cat}^{\otimes})^{\times} \cong \text{Mon}_{\text{Comm}}(\text{Cat})^{\times} \\ & \searrow \text{Span}(-) & \downarrow G \\ & & \text{Cat}^{\times} \end{array}$$

which induces the following commutative diagram:

$$\begin{array}{ccccc}
\text{Cat}^{pb} & \xrightarrow{\text{Span}_1(-)^\otimes} & & & \\
\uparrow \text{ev}_{(1)} & & & & \\
\text{CAlg}(\text{Cat}^{pb}) & \xrightarrow{\text{CAlg}(\text{Span}_1(-)^\otimes)} & \text{CAlg}(\text{Mon}_{\text{Comm}}(\text{Cat})) \cong \text{CAlg}(\text{CAlg}(\text{Cat})) & \xrightarrow{\text{ev}_{(1)}, \cong} & \text{Mon}_{\text{Comm}}(\text{Cat}) \\
& \searrow \text{CAlg}(\text{Span}_1(-)) & \downarrow \cong, \text{CAlg}(G) & & \downarrow G \\
& & \text{CAlg}(\text{Cat}) & \xrightarrow{\text{ev}_{(1)}} & \text{Cat}
\end{array}$$

Thus, given a commutative algebra object $\mathcal{C}$ in Cat^{pb} , the induced commutative algebra object $\mathcal{C} \xrightarrow{*} \text{CAlg}(\text{Cat}^{pb}) \xrightarrow{\text{Span}_1(-)} \text{CAlg}(\text{Cat})$ corresponds to one symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$. On the other hand, if we forget the commutative algebra structure on $\mathcal{C}$ and regard it as an object in Cat^{pb} , by the construction $\text{Span}_1(-)^\otimes$, there is another symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$. These symmetric monoidal structures coincide since $\text{ev}_{(1)} : \text{CAlg}(\text{CAlg}(\text{Cat})) \rightarrow \text{CAlg}(\text{Cat})$ is an equivalence.

2.3 $B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons (\text{Cat}_n)_{*/} : \Omega$

In this part, we review the symmetric monoidal adjunction

$$B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons \text{Cat}_{n*/} : \Omega$$

established in [12, remark 10.6] and [10, theorem 6.3.2], we show that $\Omega(\text{Span}_n(\mathcal{C})) \cong \text{Span}_{n-1}(\mathcal{C})$, and the induced symmetric monoidal structure on $\Omega(\text{Span}_2(\mathcal{C})) \cong \text{Span}_1(\mathcal{C})$ coincides with the symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$ in [construction 2.2.1](#).

Notation 2.3.1. Let $\text{Seg}(\text{Cat}_{n-1})$ denote the full subcategory of $\text{Cat}(\text{Cat}_{n-1})$ consisting of those functors $F : \Delta^{op} \rightarrow \text{Cat}_{n-1}$ satisfying $F([0]) \in \text{Spc} \subset \text{Cat}_{n-1}$.

Using the Segal category model, there is a functor

$$(*) \quad \text{Alg}(\text{Cat}_{n-1}) \hookrightarrow \text{Seg}(\text{Cat}_{n-1}) \xrightarrow{L} \text{Cat}_n$$

where L is the localization functor defined in [12, theorem 7.6].

On the other hand, in the model of (∞, n) category by enriched categories using categorical algebras, there is a functor defined in [10, definition 6.3.1]:

$$(**) \quad \text{Alg}(\text{Cat}_{n-1}) \hookrightarrow \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \rightarrow \text{Cat}^{\text{Cat}_{n-1}}$$

where the second functor is the localization functor.

By the comparison of the two models, established in [11], there is a commutative diagram:

$$\begin{array}{ccc}
\text{Cat}^{\text{Cat}_{n-1}} & \hookrightarrow & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
\uparrow \cong & & \uparrow \cong \\
\text{Cat}_n & \hookrightarrow & \text{Seg}(\text{Cat}_{n-1})
\end{array}$$

, and by taking the left adjoints, we get

$$\begin{array}{ccc}
 \text{Cat}^{\text{Cat}_{n-1}} & \xleftarrow{\text{localization}} & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
 \downarrow \cong & & \downarrow \cong \\
 \text{Cat}_n & \xleftarrow{\text{localization}} & \text{Seg}(\text{Cat}_{n-1})
 \end{array}$$

under the equivalence of $\text{Cat}^{\text{Cat}_{n-1}} \cong \text{Cat}_n$, the above two functors (*) and (**) from $\text{Alg}(\text{Cat}_{n-1}) \rightarrow (\text{Cat}_n)_{*/}$ coincide, and we denote it as

$$B : \text{Alg}(\text{Cat}_n) \rightarrow (\text{Cat}_n)_{*/}$$

Lemma 2.3.2. (1) *B is fully faithful.*

(2) *B is a symmetric monoidal functor, with $\text{Alg}(\text{Cat}_n)$ and $(\text{Cat}_n)_{*/} \cong \text{Alg}_{\mathbb{E}_0}(\text{Cat}_n)$ equipped with both the Cartesian symmetric monoidal structure.*

(3) *The functor B admits a right adjoint Ω .*

Proof. This is [10, theorem 6.3.2], combined with the fact that the symmetric monoidal structure on $\text{Cat}^{\text{Cat}_{n-1}}$ is Cartesian by [36, proposition 2.5.8] $\square$

Remark 2.3.3. Moreover, one can describe the right adjoint Ω explicitly: Consider $\text{Seg}(\text{Cat}_{n-1}) \cong \text{Alg}_{\text{cat}}(\text{Cat}_{n-1})$:

- There is a commutative diagram

$$\begin{array}{ccc}
 \text{Seg}(\text{Cat}_{n-1}) & \xrightarrow{\cong} & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
 \searrow \text{ev}_{[0]} & & \swarrow p \\
 & \text{Spc} &
 \end{array}$$

where $\text{ev}_{[0]}$ is induced by $[0] \rightarrow \Delta^{op}$, and p is the functor taking the underlying space S of a categorical algebra $\Delta_S^{op} \rightarrow \text{Cat}_{n-1}^{\times}$. Moreover, the above equivalence $\text{Seg}(\text{Cat}_{n-1}) \rightarrow \text{Alg}_{\text{cat}}(\text{Cat}_{n-1})$ is a morphism between Cartesian fibrations over Spc .

- By [12, lemma 6.33 & theorem 6.3.2], the above adjunction $B \dashv \Omega$ corresponds to the biCartesian fibration

$$\begin{array}{ccc}
 \square & \longrightarrow & \text{Seg}(\text{Cat}_{n-1})_{*/} \\
 \downarrow \lrcorner & & \downarrow \\
 \text{Spc}_{*/}^{\triangleleft} & \longrightarrow & \text{Spc}_{*/} \\
 \downarrow & & \\
 [1] & &
 \end{array}$$

- Thus $\Omega(X)$ computes the Cartesian lifting of X over $* \rightarrow X([0])$ for $X \in \text{Seg}(\text{Cat}_{n-1})$.

- Noticing that $ev_{[0]} : \text{Seg}(\text{Cat}_{n-1}) \subset \text{Fun}(\Delta^{op}, \text{Cat}_{n-1}) \rightarrow \text{Cat}_{n-1}$ admits a right adjoint R by taking the Čech nerve of an object to the final object, then by [13, corollary 4.5.2], we can compute the Cartesian lifting of X over the pointed map $* \rightarrow X[0]$ through the pullback diagram

$$\begin{array}{ccc} \Omega(X) & \xrightarrow{\quad} & X \\ \downarrow & \lrcorner & \downarrow \eta \\ \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(X) = X([0])^{\times \bullet + 1} \end{array}$$

where η is the coevaluation map.

- So the underlying object $\Omega(X)[1]$ of $\Omega(X)$ is equivalent to the pullback $X[1] \times_{X([0])^{\times 2}} *$.
- The algebra object structure on $\Omega(X)$ is induced by the composition law for $X([1])$: for the morphism $0, 2 : [1] \rightarrow [2] \in \Delta$, there is a composition map $X([2]) \cong X([1]) \times_{X([0])} X([1]) \rightarrow X([1])$ that restricts to the multiplication map for the algebra object $\Omega(X)[1] \in \text{Cat}_{n-1}$.
- The functoriality of the right adjoint Ω can also be described using the Cartesian lifting: given $f : X \rightarrow Y$, the induced morphism between $\Omega(f) : \Omega(X) \rightarrow \Omega(Y)$ is induced by the pullbacks:

$$\begin{array}{ccccc} \Omega(Y) & \xrightarrow{\quad} & Y & & \\ \downarrow & \searrow & \downarrow & \searrow f & \\ \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(Y) & & \\ & \searrow \Omega(f) & \downarrow & \searrow & \\ & & \Omega(X) & \xrightarrow{\quad} & X \\ & & \downarrow & \lrcorner & \downarrow \eta \\ & & \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(X) = X([0])^{\times \bullet + 1} \end{array}$$

Lemma 2.3.4. For $\mathcal{C} \in \text{Cat}^{pb}$,

- (1) For, $n \geq 1$, consider the pointed map $* \rightarrow \text{Span}_n(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$, $\Omega(\text{Span}_n(\mathcal{C})) \cong \text{Span}_{n-1}(\mathcal{C})$
- (2) For $n \geq 2$, consider the pointed map $* \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$, $\Omega(\text{Span}_{n-1}^+(\mathcal{C})) \cong \text{Span}_{n-2}^+(\mathcal{C})$.

Proof. (1) is by [12, proposition 8.3], and (2) is the same by replacing $\overline{\text{SPAN}}_n$ with $\overline{\text{SPAN}}_{n-1}^+$, SPAN_n with SPAN_{n-1}^+ , and Span_n with Span_{n-1}^+ . $\square$

Noticing that the left adjoint B in the adjunction $B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons (\text{Cat}_n)_{*/} : \Omega$ is symmetric monoidal, thus Ω can be enhanced into a lax symmetric monoidal functor. Given the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$, the pointed map $* \rightarrow \text{Span}_2(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$ is the unit map, thus it defines an object in $\text{CAlg}((\text{Cat}_n)_{*/})$ and induces a symmetric monoidal structure on $\Omega(\text{Span}_2(\mathcal{C}) \cong \text{Span}_1(\mathcal{C}))$.

To distinguish this symmetric monoidal structure from the one acquired in the last subsection (which is through the functoriality of the functor $\text{Span}_1(-)$), we denote the symmetric monoidal category $\text{Span}_1(\mathcal{C})^{\otimes'}$ as the one from the lax symmetric monoidal functor Ω , and $\text{Span}_1(\mathcal{C})^{\otimes}$ as the one from the functoriality of the finite-limit preserving functor $\text{Span}_1(-) : \text{Cat}^{bb} \rightarrow \text{Cat}_1$. Likewise, let $\text{Span}_1^+(\mathcal{C})^{\otimes'}$ denote the symmetric monoidal structure on $\Omega(\text{Span}_2^+(\mathcal{C}))$ acquired from the lax symmetric monoidal functor Ω , and $\text{Span}_1^+(\mathcal{C})^{\otimes}$ as the one from the functoriality of the functor $\text{Span}_1^+(-)$.

Proposition 2.3.5. *There is a symmetric monoidal equivalence of symmetric monoidal categories $\text{Span}_1(\mathcal{C})^{\otimes} \cong \text{Span}_1(\mathcal{C})^{\otimes'}$ and $\text{Span}_1^+(\mathcal{C})^{\otimes} \cong \text{Span}_1^+(\mathcal{C})^{\otimes'}$.*

Proof. • Input the following morphisms in $\text{CAlg}((\text{Cat}_3)_{*/})$

$$\text{Span}_1^+(\mathcal{C})^{\otimes} \rightarrow \text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \text{Span}_2^+(\mathcal{C})^{\otimes}$$

. By applying the lax symmetric monoidal functor $\Omega : (\text{Cat}_3)_{*/} \rightarrow \text{Alg}(\text{Cat}_2)$, there is an induced chain of morphisms in $\text{CAlg}(\text{Cat}_2)$:

$$\mathcal{C}^{\otimes'} \xrightarrow{i} \text{Span}_1(\mathcal{C})^{\otimes'} \xrightarrow{i^+} \text{Span}_1^+(\mathcal{C})^{\otimes'}$$

.

- $\mathcal{C}^{\otimes'}$ is equipped with the Cartesian symmetric monoidal structure, noting that the algebra object structure on $\mathcal{C}' = \Omega(\text{Span}_1^+(\mathcal{C})) \cong \underline{\text{End}}_{\text{Span}_1^+(\mathcal{C})}(\ast)$ is by composition. And the final object $\ast$ corresponds to the identity map $\ast = \ast \ast \ast$. Given $\ast \leftarrow x \rightarrow \ast, \ast \leftarrow y \rightarrow \ast \in \Omega(\text{Span}_1^+(\mathcal{C}))$, the composition map gives us $x \circ y = (\ast \leftarrow x \times y \rightarrow \ast)$, and the induced morphisms to $x \circ \ast = x$ and $\ast \circ y = y$ form a product diagram in $\mathcal{C} = \Omega \text{Span}_1^+(\mathcal{C})$.
- Thus, $\mathcal{C}^{\otimes'} \cong \mathcal{C}^{\otimes}$ are equivalent as symmetric monoidal categories, both with Cartesian symmetric monoidal structures.
- By the [remark 2.2.4](#), the inclusion $\iota_2^+ \circ \iota_1 : \mathcal{C} \rightarrow \text{Span}_1^+(\mathcal{C})$ has the universal property established in [\[35, corollary 1.2.2\]](#); thus, there is an induced morphism $\psi : \text{Span}_1^+(\mathcal{C})^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ making the following diagram commute in $\text{CAlg}(\text{Cat}_2)$:

$$\begin{array}{ccccc} \mathcal{C}^{\otimes} & \xrightarrow{i} & \text{Span}_1(\mathcal{C})^{\otimes'} & \xrightarrow{i^+} & \text{Span}_1^+(\mathcal{C})^{\otimes'} \\ \downarrow \iota_0 & & & \nearrow \psi & \\ \text{Span}_1(\mathcal{C})^{\otimes} & & & & \\ \downarrow \iota_1^+ & & & & \\ \text{Span}_1^+(\mathcal{C})^{\otimes} & & & & \end{array}$$

- By the descriptions of the functor $\Omega(-)$ at objects and morphisms in the [remark 2.3.3](#), we can see that at the level of underlying $(\infty, 2)$ categories, the above inclusion $i^+ \circ i : \mathcal{C}^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ is equivalent to the inclusion $\iota_1^+ \iota_0$, which corresponds to the non-full

subcategory of $\text{Span}_1^+(\mathcal{C})$ consisting of all the objects and 1-morphisms in the form of a span with the left leg being an equivalence and invertible 2-morphisms; thus, it also has the same universal property, and this induces an inverse functor ψ^{-1} of ψ at the level of underlying $(\infty, 2)$ categories; thus, ψ is a symmetric monoidal $(\infty, 2)$ equivalence.

- Since $i^+ : \text{Span}_1(\mathcal{C})^{\otimes'} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ exhibits $\text{Span}_1(\mathcal{C})^{\otimes'}$ as the underlying $(\infty, 1)$ -category of $\text{Span}_1^+(\mathcal{C})^{\otimes'}$, the map $\psi \circ i_1^+ : \text{Span}_1(\mathcal{C})^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ factors through $i^+ : \text{Span}_1(\mathcal{C})^{\otimes'} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ and the factorization $\text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C})^{\otimes'}$ is an equivalence at the level of underlying $(\infty, 1)$ categories; thus, it is a symmetric monoidal equivalence.

□

Corollary 2.3.6. *Consider the composition of adjunctions:*

$$B : \text{CAlg}(\text{Cat}) \rightleftharpoons \text{CAlg}(\text{Cat}_2) : \Omega, B : \text{CAlg}(\text{Cat}_2) \rightleftharpoons \text{CAlg}(\text{Cat}_3) : \Omega, \dots, B : \text{CAlg}(\text{Cat}_{n-1}) \rightleftharpoons \text{CAlg}(\text{Cat}_n) : \Omega$$

, then for $\text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$, $\Omega^{n-1}(\text{Span}_n(\mathcal{C})) \cong \text{Span}_1(\mathcal{C})$ in $\text{CAlg}(\text{Cat})$.

Proof. • Again we consider $\text{Span}_{n-1}^+(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}) \rightarrow \text{Span}_n^+(\mathcal{C})$ in $\text{CAlg}(\text{Cat}_{n+1})$, then by applying $\Omega^{n-1}(-)$, we get

$$\mathcal{C}' \rightarrow \text{Span}_1(\mathcal{C})' \rightarrow \text{Span}_1^+(\mathcal{C})'$$

where $\mathcal{C}'$, $\text{Span}_1(\mathcal{C})'$ and $\text{Span}_1^+(\mathcal{C})'$ are the same as $\mathcal{C}$, $\text{Span}_1(\mathcal{C})$ and $\text{Span}_1^+(\mathcal{C})$ at the level of underlying $(\infty, 2)$ categories but with possibly different symmetric monoidal structures induced by the lax symmetric monoidal functor $\Omega^{n-1}(-)$.

- Again one can check that $\mathcal{C}'$ is equipped with the Cartesian symmetric monoidal structure, and thus by the universal property of $\mathcal{C} \rightarrow \text{Span}_1^+(\mathcal{C})$, there is a symmetric monoidal $(\infty, 2)$ equivalence $\psi : \text{Span}_1^+(\mathcal{C}) \rightarrow \text{Span}_1^+(\mathcal{C})'$ which restricts to a symmetric monoidal $(\infty, 1)$ equivalence $\text{Span}_1(\mathcal{C}) \cong \text{Span}_1(\mathcal{C})' = \Omega^{n-1}(\text{Span}_n(\mathcal{C}))$.

□

2.4 $\text{LMod}_{B_A}(\text{Cat}^{\mathcal{M}}) \cong \text{Cat}^{\text{LMod}_A(\mathcal{M})}$

In this subsection, we review the equivalence $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \cong \text{LMod}_{B_A}(\text{Cat}^{\mathcal{M}})$ established in [37, proposition 2.2].

Given a presentably symmetric monoidal ∞ -category $\mathcal{M}$, consider $A \in \text{CAlg}(\mathcal{M})$, then there is a symmetric monoidal adjunction

$$L = (-) \otimes A : \mathcal{M} \rightleftharpoons \text{LMod}_A(\mathcal{M}) : R = \text{forget}$$

.

By [10, proposition 5.7.1], there is a functor

$$\text{Cat}^{(-)} : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$$

classified by the coCartesian fibration $\text{Enr} \rightarrow \text{Alg}(\text{Cat})$, which is a subfunctor of the functor

$$\text{Alg}_{\text{cat}}(-) : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$$

,classified by the coCartesian fibration $\text{Alg}_{\text{cat}} \rightarrow \text{Alg}(\text{Cat})$. Moreover, there is an adjunction over $\text{Alg}(\text{Cat})$:

$$\text{loc} : \text{Alg}_{\text{cat}} \rightleftarrows \text{Enr} : i$$

where loc is the localization functor taking a categorical algebra in $\mathcal{V} \in \text{Alg}(\text{Cat})$ to its completion as an $\mathcal{V}$ -enriched category.

By [15, remark 7.11], one can enhance $\text{Enr} \subset \text{Alg}_{\text{cat}} \rightarrow \text{Alg}(\text{Cat})$ into a symmetric monoidal coCartesian fibration $\text{Enr}^\times \subset \text{Alg}^\times \rightarrow \text{Alg}(\text{Cat})^\times$. So one can deduce that:

- The functor $\text{Cat}^{(-)} : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$ can be enhanced into a lax symmetric monoidal functor. Especially, given a symmetric monoidal functor $L : \mathcal{M} \rightarrow \text{LMod}_A(\mathcal{M})$, $L_* : \text{Cat}^\mathcal{M} \rightarrow \text{Cat}^{\text{LMod}_A(\mathcal{M})}$ is also symmetric monoidal. Thus, there is an induced symmetric monoidal adjunction by [10, proposition 5.7.17]

$$L_* : \text{Cat}^\mathcal{M} \rightleftarrows \text{Cat}^{\text{LMod}_A(\mathcal{M})} : R_*$$

,

- Given $\mathcal{V} \in \text{CAlg}(\text{Cat})$, the localization adjunction

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightleftarrows \text{Cat}^\mathcal{V} : i_\mathcal{V}$$

can be enhanced into a symmetric monoidal adjunction, especially, for any $x \in \text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{V}))$, the local equivalence $x \rightarrow \text{loc}_\mathcal{V}(x)$ lands in $\text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{M}))$.

Lemma 2.4.1. *Given a symmetric monoidal adjunction*

$$L : \mathcal{M} \rightleftarrows \mathcal{N} : R$$

, consider the induced symmetric monoidal adjunction

$$L_* : \text{Cat}^\mathcal{M} \rightleftarrows \text{Cat}^\mathcal{N} : R_*$$

, then there is an equivalence $R_* 1_\mathcal{N} \cong \text{BR}(1_\mathcal{M})$ in $\text{CAlg}(\text{Cat}^\mathcal{N})$ where $\text{BR}(1_\mathcal{N})$ is the image of $R(1_\mathcal{N}) \in \text{CAlg}(\mathcal{M})$ under the loop functor $\text{B} : \text{CAlg}(\text{Alg}(\mathcal{M})) \rightarrow \text{CAlg}(\text{Cat}^\mathcal{M})$.

Proof. • By definition, the symmetric monoidal functor B is the composition of the following symmetric monoidal functors:

$$\text{B} : \text{Alg}(\mathcal{N}) \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{N})_{1_\mathcal{N}/} \xrightarrow{L_\mathcal{N}} \text{Cat}_{1_{\text{Cat}^\mathcal{N}}/}^\mathcal{N}$$

where $1_\mathcal{N}$ is the unit in $\text{Alg}(\mathcal{N}) \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{N})$ and $1_{\text{Cat}^\mathcal{N}} = \text{loc}_\mathcal{N}(1_\mathcal{N})$ is its localization (they may differ when the unit $1_\mathcal{N}$ has a nontrivial endomorphism space).

- For $a \in \text{CAlg}(\text{Alg}(\mathcal{N}))$, denote its image in $\text{Alg}_{\text{cat}}(\mathcal{N})_{1_\mathcal{N}/}$ as $\text{B}^{\text{pre}} a$. Then, there is a morphism $\text{B}^{\text{pre}} 1_\mathcal{N} \rightarrow \text{B} 1_\mathcal{N}$ in $\text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{N}))$ by the symmetric monoidal adjunction $\text{loc}_\mathcal{N} \dashv i_\mathcal{N}$.

- By taking the image of $B^{pre} 1_{\mathcal{N}} \rightarrow B 1_{\mathcal{N}}$ under the functor $\text{CAlg}(R_*) : \text{CAlg}(\text{Alg}_{cat}(\mathcal{N})) \rightarrow \text{CAlg}(\text{Alg}_{cat}(\mathcal{M}))$, we then get a morphism

$$R_* B^{pre} 1_{\mathcal{N}} \cong B^{pre} R(1_{\mathcal{N}}) \rightarrow R_* B 1_{\mathcal{N}}$$

- Especially, by [10, proposition 5.7.17], $R_* B 1_{\mathcal{N}}$ is complete, and the above morphism is fully faithful and essentially surjective by [10, lemma 5.7.5].
- Thus, by the symmetric monoidal adjunction of $L_{\mathcal{M}} \dashv i_{\mathcal{M}}$, the induced morphism $B R(1_{\mathcal{N}}) \cong \text{loc}_{\mathcal{M}}(B^{pre} R(1_{\mathcal{N}})) \rightarrow R_* B 1_{\mathcal{N}}$ is an equivalence in $\text{CAlg}(\text{Cat}^{\mathcal{M}})$.

□

Now consider the lax symmetric monoidal right adjoint $R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$, there is an induced lax symmetric monoidal functor

$$R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{LMod}_{R_* 1_{\text{LMod}_A(\mathcal{M})}}(\text{Cat}^{\mathcal{M}}) \cong \text{LMod}_{B A}(\text{Cat}^{\mathcal{M}})$$

where $B A$ is the image of $A \in \text{CAlg}(\mathcal{M})$ under the symmetric monoidal functor $B : \text{Alg}(\mathcal{M}) \rightarrow (\text{Cat}^{\mathcal{M}})_*/$

Moreover, one can show that

Corollary 2.4.2. *In the above context, $R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{LMod}_{B A}(\text{Cat}^{\mathcal{M}})$ is a symmetric monoidal equivalence.*

Proof. This is by [37, proposition 2.2] □

Remark 2.4.3. For further purpose, one can show that the above symmetric monoidal equivalence is functorial:

- Given a morphism between commutative algebras $f : A \rightarrow B$ in $\text{CAlg}(\mathcal{M})$, there is a

$$\text{commutative diagram in } \text{CAlg}(\widehat{\text{Cat}}): \begin{array}{ccc} \text{LMod}_B(\mathcal{M}) & \xleftarrow{(-) \otimes B} & \mathcal{M} \\ (-) \otimes_{AB} \uparrow & & \parallel \\ \text{LMod}_A(\mathcal{M}) & \xleftarrow{(-) \otimes A} & \mathcal{M} \end{array}, \text{ which gives rise to a}$$

$$\text{commutative morphsim in } \text{CAlg}(\widehat{\text{Cat}}): \begin{array}{ccc} \text{Cat}^{\text{LMod}_B(\mathcal{M})} & \xleftarrow{((-) \otimes B)_*} & \text{Cat}^{\mathcal{M}} \\ ((-) \otimes_{AB})_* \uparrow & & \parallel \\ \text{Cat}^{\text{LMod}_A(\mathcal{M})} & \xleftarrow{((-) \otimes A)_*} & \text{Cat}^{\mathcal{M}} \end{array}.$$

- By passing to the right adjoints, we get a morphism in Op :

$$\begin{array}{ccc} \text{Cat}^{\text{LMod}_B(\mathcal{M})} & \xrightarrow{(F_{B/1})_*} & \text{Cat}^{\mathcal{M}} \\ (F_{B/A})_* \downarrow & & \parallel \\ \text{Cat}^{\text{LMod}_A(\mathcal{M})} & \xrightarrow{(F_{A/1})_*} & \text{Cat}^{\mathcal{M}} \end{array}$$

where $(F_{B/A})_* : \text{Cat}^{\text{LMod}_B(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$ induced by the transfer of enrichment along the forgetful functor $F_{B/A} : \text{LMod}_B(\mathcal{M}) \rightarrow \mathcal{M}$, likewise for $F_{B/1}$ and $F_{A/1}$.

- By taking the induced morphisms between left module categories of the above diagram, we get:

$$\begin{array}{ccc}
\mathrm{LMod}_{1_{\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})}}}(\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})}) \cong \mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})} & \xrightarrow{(F_{B/1})_*} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow (F_{B/A})_* & & \downarrow \cong \\
\mathrm{LMod}_B(\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})}) & \xrightarrow{(F_{A/1})_*} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

- Now consider the symmetric monoidal equivalence in [corollary 2.4.2](#): $\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} \rightarrow \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})$, by taking the induced left module categories, there is a commutative diagram:

$$\begin{array}{ccc}
\mathrm{LMod}_B(\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})}) & \xrightarrow{(F_{A/1})_*} & \mathrm{LMod}_B(\mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})) \cong \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow \text{forget} & & \downarrow B A \rightarrow B B \\
\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

- By combining the above two commutative diagrams together, there is a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow (F_{B/A})_* & & \downarrow B A \rightarrow B B \\
\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

where the left vertical morphism is induced by the transfer of enrichment along the forgetful functor $\mathrm{LMod}_B(\mathcal{M}) \rightarrow \mathrm{LMod}_A(\mathcal{M})$, and the right vertical morphism is induced by the forgetful functor $\mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \rightarrow \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})$ along $B A \rightarrow B B$, and the horizontal morphisms are equivalences by [corollary 2.4.2](#).

2.5 $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \hookrightarrow \mathrm{Cat}^{\mathrm{Span}_1(\mathcal{C})}$

This subsection is an expansion of the [\[37, section 2\]](#).

By [\[15, theorem 7.21\]](#), there is a lax symmetric monoidal functor

$$\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat}) \rightarrow \mathrm{Cat}^{\mathcal{P}(\mathrm{Span}_1(\mathcal{C}))}$$

which is a non-full subcategory inclusion.

Consider the full subcategory $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \subset \mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})$ consisting of those left $\mathrm{Span}_1(\mathcal{C})$ -modules with closed action, then one can show that:

Lemma 2.5.1. *The embedding $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \hookrightarrow \mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})$ is closed under tensor product $\otimes_{\mathrm{Span}_1(\mathcal{C})}$.*

Proof. It suffices to show that for $\mathcal{M}, \mathcal{N} \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$, the relative tensor product $\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ is again a closed $\text{Span}_1(\mathcal{C})$ -module.

- Noticing that the canonical map $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ is surjective at the level of underlying spaces since it's the last term in the bar construction, thus it suffices to show that for any $m \in \mathcal{M}, n \in \mathcal{N}, m \boxtimes n \in \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$, the morphism $\text{Span}_1(\mathcal{C}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}) \rightarrow \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ induced by $\text{Span}_1(\mathcal{C}) \rightarrow \mathcal{M}$ specifying the object m and $\text{Span}_1(\mathcal{C}) \rightarrow \mathcal{N}$ specifying the object n , admits a right adjoint $\underline{\text{Hom}}_{\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}}(m \boxtimes n, -)$.
- Noticing that the symmetric monoidal functor $\text{Cat} \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ makes $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ into a symmetric monoidal $(\infty, 2)$ category, thus the tensor product $\otimes_{\text{Span}_1(\mathcal{C})}$ preserves adjunction. And it remains to show that the right adjoints

$$\underline{\text{Hom}}_{\mathcal{M}}(m, -) : \mathcal{M} \rightarrow \text{Span}_1(\mathcal{C}), \underline{\text{Hom}}_{\mathcal{N}}(n, -) : \mathcal{N} \rightarrow \text{Span}_1(\mathcal{C})$$

are $\text{Span}_1(\mathcal{C})$ -linear functors, which is true by noting that every object in $\text{Span}_1(\mathcal{C})$ has a dual: for any $a \in \text{Span}_1(\mathcal{C})$,

$$\text{Hom}_{\mathcal{M}}(m, a \otimes (-)) \cong \text{Hom}_{\mathcal{M}}(m \otimes a^\vee, (-)) \cong \underline{\text{Hom}}_{\text{Span}_1(\mathcal{C})}(a^\vee \underline{\text{Hom}}_{\mathcal{M}}(m, (-)) \cong a \otimes \underline{\text{Hom}}_{\mathcal{M}}(m, (-))$$

□

Remark 2.5.2. Especially, by the above remark, we can see that

$$\underline{\text{Hom}}_{\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}}(m \boxtimes n, x \boxtimes y) \cong \underline{\text{Hom}}_{\mathcal{M}}(m, x) \boxtimes \underline{\text{Hom}}_{\mathcal{N}}(n, y) \in \text{Span}_1(\mathcal{C}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}) \cong \underline{\text{Hom}}_{\mathcal{M}}(m, x) \times \underline{\text{Hom}}_{\mathcal{N}}(n, y)$$

.

Thus, we can enhance $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$ as a symmetric monoidal full subcategory of $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, and we get a lax symmetric monoidal functor

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

.

Noticing that the symmetric monoidal functor $(\mathcal{L}_{\text{Span}_1(\mathcal{C})})_* : \text{Cat}^{\text{Span}_1(\mathcal{C})} \hookrightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ is fully faithful by [16, corollary 4.74], thus the above inclusion factors as

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

.

Moreover, one can show that:

Lemma 2.5.3. *The inclusion $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$ is a fully faithful embedding.*

Proof. • We work in the model of enriched category by weakly left tensored categories, and it suffices to show that any lax $\text{Span}_1(\mathcal{C})$ left module morphism between $\mathcal{M}, \mathcal{N} \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$ is also a strict left module morphism: that is, an operad morphism $F : (\text{Span}_1(\mathcal{C}), \mathcal{M}) \rightarrow (\text{Span}_1(\mathcal{C}), \mathcal{N})$ over $\text{LM}^\otimes$ which is identity over $\text{Ass}^\otimes \hookrightarrow \text{LM}^\otimes$ will automatically preserve the coCartesian liftings over the active morphism in $\text{LM}^\otimes$.

- It suffices to show that for any $a \in \text{Span}_1(\mathcal{C})$, $m \in \mathcal{M}$, the induced morphism

$$\rho : a \otimes F(m) \rightarrow F(a \otimes m)$$

is an equivalence.

- We construct the inverse map of ρ $\tau : F(a \otimes m) \rightarrow a \otimes F(m)$ by the dual map $a^\vee \otimes F(a \otimes m) \rightarrow F(a^\vee \otimes a \otimes m) \xrightarrow{F(\text{coev}) \otimes \text{id}} F(m)$, and it remains to show that the following compositions are equivalent to identity maps

$$\begin{aligned} a \otimes F(m) &\xrightarrow{\rho} F(a \otimes m) \xrightarrow{\tau} a \otimes F(m) \\ F(a \otimes m) &\xrightarrow{\tau} a \otimes F(m) \xrightarrow{\rho} F(a \otimes m) \end{aligned}$$

.

- For the first composition of morphisms, notice the following commutative diagrams:

$$\begin{array}{ccc} a \otimes F(x) & \xrightarrow{\text{coev}} & (a \otimes a^\vee) \otimes a \otimes F(x) \\ \downarrow \rho & & \downarrow \\ F(a \otimes x) & \longrightarrow & (a \otimes a^\vee) \otimes F(a \otimes x) \end{array}$$

and

$$\begin{array}{ccc} (a \otimes a^\vee) \otimes a \otimes F(x) & \longrightarrow & a \otimes 1 \otimes F(x) \\ \downarrow \cong & \nearrow \text{ev} & \downarrow \cong \\ a \otimes (a^\vee \otimes a) \otimes F(x) & & a \otimes F(1 \otimes x) \\ \downarrow & \searrow \text{ev} & \\ a \otimes F(a^\vee \otimes a \otimes x) & \longrightarrow & a \otimes F(1 \otimes x) \end{array}$$

, which combine into

$$\begin{array}{ccccccc} a \otimes F(x) & \xrightarrow{\text{coev}} & (a \otimes a^\vee) \otimes a \otimes F(x) & & & & \\ \downarrow \rho & & \downarrow & & \searrow \text{ev} & & \\ F(a \otimes x) & \longrightarrow & (a \otimes a^\vee) \otimes F(a \otimes x) & \longrightarrow & a \otimes F(a^\vee \otimes a \otimes x) & \longrightarrow & a \otimes F(x) \\ & & & \nearrow \tau & & & \end{array}$$

- For the second composition, consider the following commutative diagrams:

$$\begin{array}{ccc} 1 \otimes F(a \otimes x) & \xrightarrow{\cong} & F(1 \otimes a \otimes x) \\ \text{coev} \downarrow & & \text{coev} \downarrow \\ (a \otimes a^\vee) \otimes F(a \otimes x) & \longrightarrow & F((a \otimes a^\vee) \otimes a \otimes x) \end{array}$$

and

$$\begin{array}{ccc} a \otimes F(a^\vee \otimes a \otimes x) & \xrightarrow{\text{ev}} & a \otimes F(x) \\ \downarrow & & \downarrow \rho \\ F(a \otimes (a^\vee \otimes a) \otimes x) & \xrightarrow{\text{ev}} & F(a \otimes x) \end{array}$$

which combine into

$$\begin{array}{ccccccc}
 & & a \otimes F(a^\vee \otimes a \otimes x) & \xrightarrow{coev} & a \otimes F(1 \otimes x) & & \\
 & \nearrow & \downarrow & & \searrow \rho & & \\
 (a \otimes a^\vee) \otimes F(a \otimes x) & \xrightarrow{\quad} & F((a \otimes a^\vee) \otimes a \otimes x) & \xrightarrow{\cong} & F(a \otimes (a^\vee \otimes a) \otimes x) & \xrightarrow{coev} & F(a \otimes x) \\
 \uparrow ev & & \nearrow ev & & & & \\
 1 \otimes F(a \otimes x) & \cong & F(1 \otimes a \otimes x) & & & &
 \end{array}$$

where the compositions of the morphisms from $1 \otimes F(a \otimes x)$ at the bottom to $F(a \otimes x)$ at the right hand side along the upper path give us $\rho \circ \tau$, and the compositions along the lower path give us id .

□

2.6 $\text{Span}_n^{enr}(\mathcal{C}) \in n \text{Cat}^{\text{Span}_1(\mathcal{C})}$

The following construction is [37, section 2]:

Notation 2.6.1. Given a symmetric monoidal category $\mathcal{V}$, $\text{Cat}^\mathcal{V}$ is again symmetric monoidal,

- (1) We set $k \text{Cat}^\mathcal{V}$ as $\mathcal{V}$ when $k = 0$, and $(k + 1) \text{Cat}^\mathcal{V}$ as $\text{Cat}^k \text{Cat}^\mathcal{V}$ for $k \geq 0$.
- (2) We set $B^0 = \text{id}$ and $B^{k+1}(-) = B \circ B^k(-)$ for the symmetric monoidal adjunction $B : \text{Alg}(\mathcal{V}) \rightleftarrows (\text{Cat}^\mathcal{V})_{*/} : \Omega$.

Construction 2.6.2. Now consider $B \Omega \text{Span}_2(\mathcal{C}) \cong B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}_2)$ induced by the symmetric monoidal adjunction $B \dashv \Omega$, by [proposition 2.3.5](#), $B \text{Span}_1(\mathcal{C})$ here is equipped with the commutative algebra structure induced by $\text{Span}_1(\mathcal{C}) \in \text{CAlg}(\text{Cat})$ under the symmetric monoidal functor B .

Under the equivalence $\text{CAlg}(\text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2)) \cong \text{CAlg}(\text{Cat}_2)_{B \text{Span}_1(\mathcal{C})/}$, the map $B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})$ makes $\text{Span}_2(\mathcal{C})$ into a commutative algebra object in $\text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2) \cong \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$, denoted as $\text{Span}_2^{enr}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Inductively, for $n \geq 2$, consider $B^{n-1} \text{Span}_1(\mathcal{C}) \hookrightarrow \text{Span}_n(\mathcal{C})$. This makes $\text{Span}_n(\mathcal{C})$ into a commutative algebra object in $\text{LMod}_{B^{n-1} \text{Span}_1(\mathcal{C})}(\text{Cat}_n) \cong (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$, denoted as $\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Remark 2.6.3. If we further consider the symmetric monoidal (∞, n) functors:

$$* = B^{n-1} * \rightarrow B^{n-1} \text{Span}_1(\mathcal{C}) \hookrightarrow \text{Span}_n(\mathcal{C})$$

, by [remark 2.4.3](#), the equivalence $\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \cong \text{CAlg}(\text{LMod}_{B A}(\text{Cat}^\mathcal{M})) \cong \text{CAlg}(\text{Cat}^\mathcal{M})_{B A/}$ is functorial, and we get the following commutative diagram:

$$\begin{array}{ccc}
 \text{Span}_2^{enr}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_2)_{B \text{Span}_1(\mathcal{C})/} \ni (B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})) \\
 \downarrow & & \downarrow * \rightarrow B \text{Span}_1(\mathcal{C}) \\
 \text{CAlg}(\text{Cat}^{\text{Cat}} = \text{Cat}_2) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_2)_{*/} \ni \text{Span}_2(\mathcal{C})
 \end{array}$$

especially, this implies that under the transfer of enrichment along the forgetful functor $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}$, we recover the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}_2)$. Inductively, one can show that

$$\begin{array}{ccc} \text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C})/} \ni (B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})) \\ \downarrow & & \downarrow \scriptstyle * \rightarrow B \text{Span}_1(\mathcal{C}) \\ \text{CAlg}(\text{Cat}^{\text{Cat}_{n-1}} = \text{Cat}_n) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_n)_{*/} \ni \text{Span}_n(\mathcal{C}) \end{array}$$

, thus under the transfer of enrichment along the forgetful functor $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}$, we can recover the symmetric monoidal (∞, n) category structure on $\text{Span}_n(\mathcal{C})$ from $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Now consider the fully faithful symmetric monoidal embedding $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$. We can check that

Lemma 2.6.4. $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$ further lands inside the full subcategory $\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}} \hookrightarrow \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$.

Before the proof, one need to make the following observations about how we recover the left A -module structure on the enriched Hom $\mathcal{C}(x, y)$ for a given $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\mathcal{M}})$ under the equivalence $\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \cong \text{CAlg}(\text{Cat}^{\mathcal{M}})_{B A/}$ in [corollary 2.4.2](#).

Observation 2.6.5. • Given $\mathcal{C} \in \text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})})$, there is a multiplication map for $\mathcal{C}$ in $\text{Cat}^{\text{LMod}_A(\mathcal{M})}$:

$$m_{\mathcal{C}}^A : \mathcal{C} \otimes_A \mathcal{C} \rightarrow \mathcal{C}$$

where $\mathcal{C} \otimes_A \mathcal{C}$ is the tensor product for the symmetric monoidal category $\text{Cat}^{\text{LMod}_A(\mathcal{M})}$, for objects $x, y \in \mathcal{C}$, there is an induced morphism of enriched Homs in $\text{LMod}_A(\mathcal{M})$:

$$\text{act}_{x,y}^A : (\mathcal{C} \otimes_A \mathcal{C})(1 \otimes x, 1 \otimes y) \cong \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) \rightarrow \mathcal{C}(x, y)$$

where 1 is the unit object in $\mathcal{C}$.

- Under the lax symmetric monoidal functor $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$, the image of $\mathcal{C}$ in $\text{Cat}^{\mathcal{M}}$ is again a commutative algebra object, and we denote its multiplication map in $\text{Cat}^{\mathcal{M}}$ as $m_{\mathcal{C}}$. There is a factorization for the map $m_{\mathcal{C}}$:

$$m_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \rightarrow \mathcal{C} \otimes_A \mathcal{C} \xrightarrow{m_{\mathcal{C}}^A} \mathcal{C}$$

, and there is an induced morphism of enriched Homs in $\mathcal{M}$:

$$(!) \quad \text{act}_{x,y} : \mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) \xrightarrow{s} \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) \xrightarrow{\text{act}_{x,y}^A} \mathcal{C}(x, y)$$

(without abuse of notation, we again denote the image of $\mathcal{C}(1, 1), \mathcal{C}(x, y) \in \text{LMod}_A(\mathcal{M})$ in $\mathcal{M}$ as $\mathcal{C}(1, 1)$ and $\mathcal{C}(x, y)$) where $\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y)$ is the tensor product inside the symmetric monoidal category $\mathcal{M}$, and $\mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y)$ is the tensor product inside the symmetric monoidal category $\text{LMod}_A(\text{Cat})$.

- Moreover, the map s in (!) is the canonical factorization induced by the lax symmetric monoidal structure on the functor $\mathbf{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$. To show this, we work in the model of enriched categories by categorical algebras:

(1) Noticing that for the symmetric monoidal adjunction

$$loc_{\mathcal{M}} : \mathbf{Alg}_{cat}(\mathcal{M}) \rightleftarrows \mathbf{Cat}^{\mathcal{M}} : i_{\mathcal{M}}$$

and

$$loc_{\mathbf{LMod}_A(\mathcal{M})} : \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightleftarrows \mathbf{Cat}^{\mathbf{LMod}_A(\mathcal{M})} : i_{\mathbf{LMod}_A(\mathcal{M})}$$

, the right adjoints are usually only lax symmetric monoidal, that is,

$$(!) \quad \mathcal{C} \otimes^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes \mathcal{C}, \mathcal{C} \otimes_A^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes_A \mathcal{C}$$

is not an equivalence in the category of categorical algebras (where $\otimes^{cat}$ denotes the tensor product in the symmetric monoidal category $\mathbf{Alg}_{cat}(\mathcal{M})$ and $\otimes_A^{cat}$ denotes the tensor product of symmetric monoidal category $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M}))$).

- (2) However, the above induced morphisms ((!)) is a local equivalence for $loc_{\mathcal{M}} \dashv i_{\mathcal{M}}$ by [10, proposition 5.7.14], [10, proposition 5.7.17] and [10, lemma 7.5], that is, they are enriched fully faithful and essential surjective. Consider the following diagram in $\mathbf{Alg}_{cat}(\mathcal{M})$:

$$\begin{array}{ccccc} \mathcal{C} \otimes^{cat} \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A^{cat} \mathcal{C} & \xrightarrow{m_e^{A,cat}} & \mathcal{C} \\ \downarrow & & \downarrow & & \downarrow \cong \\ \mathcal{C} \otimes \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A \mathcal{C} & \xrightarrow{m_e^A} & \mathcal{C} \\ & \searrow m_e & & & \end{array}$$

where all the vertical arrows are the localizations and the upper horizontal arrows are the factorization of the multiplication map for $\mathcal{C}$ regarded as a commutative algebra object in $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightarrow \mathbf{Alg}_{cat}(\mathcal{M})$, and the vertical arrows induce equivalences for the enriched Homs.

Thus we may assume that the morphism (!) is induced by the factorization for the multiplication map m_e^{cat} of $\mathcal{C}$ under the lax symmetric monoidal functor $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightarrow \mathbf{Alg}_{cat}(\mathcal{M})$:

$$m_e^{cat} : \mathcal{C} \otimes^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes_A^{cat} \mathcal{C} \xrightarrow{m_e^{A,cat}} \mathcal{C}$$

- (3) Noticing that this factorization map is induced by the Beck-Chevalley map by taking the vertical right adjoints of the following commutative diagram:

$$\begin{array}{ccccc} \mathbf{Alg}_{cat}(\mathcal{M}) \times \mathbf{Alg}_{cat}(\mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \mathbf{Alg}_{cat}(\mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \mathbf{Alg}_{cat}(\mathcal{M}) \\ ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\ \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \times \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\mathbf{LMod}_A(\mathcal{M})}} & \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M}) \times \mathbf{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\mathbf{LMod}_A(\mathcal{M})})_*} & \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \end{array}$$

, where $\boxtimes_{\mathcal{M}}$ sends a pair of categorical algebras $F : \mathbf{Ass}_X \rightarrow \mathcal{M}, G : \mathbf{Ass}_Y \rightarrow \mathcal{M}$ to $F \times_{\mathbf{Ass}} G : \mathbf{Ass}_{X \times Y} \rightarrow \mathcal{M} \times \mathcal{M}$, so is $\boxtimes_{\mathbf{LMod}_A(\mathcal{M})}$.

- (4) For $(\mathcal{C}, \mathcal{C}) \in \text{Alg}_{\text{cat}}(\text{LMod}_A(\mathcal{M})) \times \text{Alg}_{\text{cat}}(\text{LMod}_A(\mathcal{M}))$, let $F : \text{Ass}_X \rightarrow \text{LMod}_A(\mathcal{M})$ denote the categorical algebra $\mathcal{C}$, and $(1, x), (1, y) \in X \times X$ are objects of $(\mathcal{C}, \mathcal{C})$ (1 is the unit object), then the Beck-Chevalley map evaluated at $(\mathcal{C}, \mathcal{C})$ agrees with the Beck-Chevalley map associated to the diagram of fibers:

$$\begin{array}{ccccc}
 \text{Alg}_{\text{Ass}_X}(\mathcal{M}) \times \text{Alg}_{\text{Ass}_X}(\mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \text{Alg}_{\text{Ass}_{X \times X}}(\mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \text{Alg}_{\text{Ass}_{X \times X}}(\mathcal{M}) \\
 ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\
 \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M})) \times \text{Alg}_{\text{Ass}_X}(\text{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\text{LMod}_A(\mathcal{M})}} & \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\text{LMod}_A(\mathcal{M})})_*} & \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M}))
 \end{array}$$

- (5) By [10, section 3.4], there is a trivial generalized non-symmetric operad $\text{Ass}_X^{\text{triv}} \rightarrow \text{Ass}_X$ associated to Ass_X satisfying the property that $\text{Alg}_{\text{Ass}_X^{\text{triv}}}(\mathcal{V}) \cong \text{Fun}(X \times X, \mathcal{V})$ for any non-symmetric operad $\mathcal{V}$. Thus by taking the induced morphism $\text{Ass}_X^{\text{triv}} \rightarrow \text{Ass}_X$, the above diagram translates into

$$\begin{array}{ccccc}
 \text{Fun}(X^{\times 2}, \mathcal{M}) \times \text{Fun}(X^{\times 2}, \mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \text{Fun}(X^{\times 4}, \mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \text{Fun}(X^{\times 4}, \mathcal{M}) \\
 ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\
 \text{Fun}(X^{\times 2}, \text{LMod}_A(\mathcal{M})) \times \text{Fun}(X^{\times 2}, \text{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\text{LMod}_A(\mathcal{M})}} & \text{Fun}(X^{\times 4}, \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\text{LMod}_A(\mathcal{M})})_*} & \text{Fun}(X^{\times 4}, \text{LMod}_A(\mathcal{M}))
 \end{array}$$

- (6) The map s between enriched Homs in (!) is the map by evaluating the Beck-Chevalley map associated to the above diagram at $(1, 1) \in X \times X, (x, y) \in X \times X$, which boils down to the Beck-Chevalley map (by taking vertical right adjoints) of the diagram:

$$\begin{array}{ccc}
 \mathcal{M} \times \mathcal{M} & \xrightarrow{\text{mult}_{\mathcal{M}}} & \mathcal{M} \\
 ((-)\otimes A) \times ((-)\otimes A) \downarrow & & (-)\otimes A \downarrow \\
 \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M}) & \xrightarrow{\text{mult}_{\text{LMod}_A(\mathcal{M})}} & \text{LMod}_A(\mathcal{M})
 \end{array}$$

, which value at $(x, y) \in \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})$ is precisely the canonical map $x \times y \rightarrow x \otimes_A y$ for $x, y \in \text{LMod}_A(\mathcal{M})$, induced by the lax symmetric monoidal structure functor $\text{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$.

- Consider the structure morphism $BA \rightarrow \mathcal{C} \in (\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \rightarrow \text{CAlg}(\text{Cat}^{\mathcal{M}}))$, and the commutative diagram induced by the multiplication maps:

$$\begin{array}{ccccc}
 BA \otimes \mathcal{C} & \longrightarrow & BA \otimes_A \mathcal{C} & & \\
 \downarrow & & \downarrow & \searrow \cong & \\
 \mathcal{C} \otimes \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A \mathcal{C} & \xrightarrow{m_{\mathcal{C}}^A} & \mathcal{C} \\
 & \searrow m_{\mathcal{C}} & & & \uparrow
 \end{array}$$

where the upper horizontal map is the BA -module structure map on $\mathcal{C}$ under the equivalence $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \cong \text{LMod}_{BA}(\text{Cat}^{\mathcal{M}})$

- For any $x, y \in \mathcal{C}$, the above diagram gives rise to a the following commutative diagram:

$$\begin{array}{ccccc}
A \otimes \mathcal{C}(x, y) & \xrightarrow{s_A} & A \otimes_A \mathcal{C}(x, y) & & \\
\downarrow t \otimes \text{id} & & \downarrow t \otimes_A \text{id} & \searrow \cong & \\
\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) & \xrightarrow{s} & \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) & \xrightarrow{\text{act}_{x,y}^A} & \mathcal{C}(x, y)
\end{array}$$

where $t : A \rightarrow \mathcal{C}(1, 1)$ is induced by the structure map $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})})$, and s_A is the factorization map induced by the lax symmetric monoidal structure on $\text{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$.

- Especially, s_A recovers the left A action map for $\mathcal{C}(x, y) \in \text{LMod}_A(\mathcal{M})$. In conclusion, given a morphism of commutative algebra objects $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\mathcal{M}})$, we can recover the left A module structure on $\mathcal{C}(x, y)$ by the composition of $A \rightarrow \mathcal{C}(1, 1)$ with the map $\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) \cong (\mathcal{C} \otimes \mathcal{C})(1 \otimes x, 1 \otimes y) \rightarrow \mathcal{C}(x, y)$ induced by the multiplication map $m_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \rightarrow \mathcal{C}$.

Proof of Lemma 2.6.4. We show this by induction: for $n = 2$,

- The action map for $\text{Span}_2^{\text{enr}}(\mathcal{C}) \in \text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2)$ is induced by

$$B \text{Span}_1(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \hookrightarrow \text{Span}_2(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \xrightarrow{\text{mult}} \text{Span}_2(\mathcal{C})$$

where the first arrow is the full $(\infty, 2)$ category embedding, and $\text{mult} : \text{Span}_2(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \cong \text{Span}_2(\mathcal{C} \times \mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})$ is induced by $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$.

- Thus for $x, y \in \text{Span}_2(\mathcal{C})$, under the equivalences

$$\underline{\text{Hom}}_{\text{Span}_2(\mathcal{C} \times \mathcal{C})}((*, x), (*, y)) \cong \text{Span}_1((\mathcal{C} \times \mathcal{C})_{/(*, x \times y)}) \cong \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x \times y})$$

and

$$\underline{\text{Hom}}_{\text{Span}_2(\mathcal{C})}(* \times x, * \times y) \cong \text{Span}_1(\mathcal{C}_{x \times y})$$

, the action map

$$\text{act} : \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$$

is induced by $\times : (\mathcal{C} \times \mathcal{C})_{/(*, x \times y)} \rightarrow \mathcal{C}_{/x \times y}$.

- Thus the action is closed: for any $t \in \text{Span}_1(\mathcal{C}_{/x \times y})$, $\text{act}(-, t) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$ is induced by $(-) \times t : \mathcal{C} \rightarrow \mathcal{C}_{/x \times y}$, which admits a right adjoint by

$$\text{Span}_1(\mathcal{C}_{/x \times y}) \xrightarrow{(-) \times_{x \times y} t} \text{Span}_1(\mathcal{C}_{/t}) \xrightarrow{\text{forget}} \text{Span}_1(\mathcal{C}).$$

Assume that for $k = n \geq 2$, the morphism $B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$ makes $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$ lands in $(n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\text{closed}}}$, then for $k = n+1$:

- For any $x, y \in \mathcal{C}$, the left module structure map gives us:

$$B^n \text{Span}_1(\mathcal{C}) \times \text{Span}_{n+1}(\mathcal{C}) \rightarrow \text{Span}_{n+1}(\mathcal{C}) \times \text{Span}_{n+1}(\mathcal{C}) \cong \text{Span}_{n+1}(\mathcal{C} \times \mathcal{C}) \xrightarrow{\text{mult}} \text{Span}_{n+1}(\mathcal{C})$$

- Thus the induced action of enriched Homs is given by:

$$B^{n-1} \text{Span}_1(\mathcal{C}) \times \text{Span}_n(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_n(\mathcal{C}) \times \text{Span}_n(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$$

, similar to the case when $n = 2$, the last action map is induced by the symmetric monoidal (∞, n) functor $(-) \times (x \times y) : \text{Span}_n(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$, thus the $B^{n-1} \text{Span}_1(\mathcal{C})$ action is induced by :

$$B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y}) \hookrightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$$

where the first arrow is $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$.

- By assumption, the left $B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y})$ on $\text{Span}_n(\mathcal{C}_{/x \times y})$ makes $\text{Span}_n(\mathcal{C}_{/x \times y})$ enhance into an object $\text{Span}_n^{enr}(\mathcal{C}_{/x \times y}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})^{\text{closed}}} \hookrightarrow (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})}$, by the functoriality in [remark 2.4.3](#), there is a commutative diagram:

(□)

$$\begin{array}{ccc} \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})}) & \xrightarrow{\quad} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y})/} \\ \downarrow & & \downarrow \\ \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\quad} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C})/} \end{array}$$

where the vertical maps are induced by the forgetful functor along $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$.

Thus it remains to show that the composition map

$$\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})^{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat}) \xrightarrow{(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$$

factors through the full subcategory $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, which is true since the functor $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$ admits a right adjoint functor induced by the slice forgetful functor $\mathcal{C}_{/x \times y} \rightarrow \mathcal{C}$. $\square$

By [\[15, theorem 7.21\]](#), there is a lax symmetric monoidal functor

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

which is a non-full subcategory inclusion.

Proposition 2.6.6. *There exists a symmetric monoidal adjunction*

$$\text{loc} : \text{Cat}^{\text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))} \rightleftarrows \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} : i$$

induced by the symmetric monoidal adjunction $\text{loc} : \text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})) \rightleftarrows \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:
i by [25, 2.2.4], there thus exists a lax symmetric monoidal functor

$$\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow \text{Cat}^{\text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

. And by induction, there is a lax symmetric monoidal functor from $(n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$.

Now consider the symmetric monoidal full subcategory embedding $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, we have

Corollary 2.6.7. For $n \geq 2$ and $\mathcal{C} \in \text{Cat}^{pb}$, $\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$ lands inside $\text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}})$, thus lands inside $\text{CAlg}(n \text{Cat}^{\text{Span}_1(\mathcal{C})})$.

3 $\text{Span}_n^{enr}(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$

In this section, we assume that $n \geq 2$, and we show that there exists a lax symmetric monoidal (∞, n) functor $\text{Span}_{n-1}^{enr}(\mathcal{C}_{/(-)}) : \text{Span}_n^{enr}(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$.

3.1 $\Gamma : \mathcal{P}_{(n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}}(\text{Span}_n^{enr}(\mathcal{C})) \rightarrow (n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$

Assume $n \geq 2$ in this subsection.

By corollary 2.6.7, we have

$$\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}(n \text{Cat}^{\text{Span}_1(\mathcal{C})}) \hookrightarrow \text{CAlg}(n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))})$$

Remark 3.1.1. • For a presentably monoidal category $\mathcal{V}$, $\text{Cat}^{\mathcal{V}}$ is again presentable by [16, proposition 2.60, corollary 4.62].

- Thus for any $n \geq 1$, $n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ is presentably symmetric monoidal category.

Notation 3.1.2. Let $\mathcal{V}$ denote the presentably symmetric monoidal category $(n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$.

Consider the enriched Yoneda embedding:

$$\mathbf{y}_{\mathcal{V}} : \text{Span}_n^{enr}(\mathcal{C}) \hookrightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C}))$$

which lands in $\text{CAlg}(\text{Cat}^{\mathcal{V}})$ by [17, corollary 1.7] (The functor $\mathcal{P}_{\mathcal{V}}$ is a symmetric monoidal left adjoint of the inclusion $\text{LMod}_{\mathcal{V}}(\text{Pr}^L) \hookrightarrow \widehat{\text{Cat}}^{\mathcal{V}}$, and the unit map of the above symmetric monoidal adjunction is the enriched Yoneda embedding.)

Now consider $\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C})) \in \text{CAlg}(\text{LMod}_{\mathcal{V}}(\text{Pr}^L))$, consider the unit map

$$1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C}))$$

, then we have

Lemma 3.1.3. *The above symmetric monoidal functor*

$$1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$$

is an internal left adjoint in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$.

Proof. • First noticing that the functor 1 can be equivalently regarded as the functor

$$\mathcal{P}_{\mathcal{V}}(\text{B } 1_{\mathcal{V}}) \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$$

induced by the symmetric monoidal $\mathcal{V}$ -enriched functor

$$\text{B } 1_{\mathcal{V}} \cong 1_{\text{Cat}^{\mathcal{V}}} \rightarrow \text{Span}_n^{\text{enr}}(\mathcal{C})$$

in $\text{CAlg}(n \text{ Cat}^{\text{Span}_1(\mathcal{C})})$ by applying the enriched $\mathcal{V}$ -presheaf functor $\mathcal{P}_{\mathcal{V}}$.

- Since $\mathcal{V} \cong \mathcal{P}_{\mathcal{V}}(\text{B } 1_{\mathcal{V}})$ is $\mathcal{V}$ -atomically generated by the unit $1_{\mathcal{V}}$, and $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$ sends the atomic generator $1_{\mathcal{V}}$ to the atomic object $* \in \text{Span}_n^{\text{enr}}(\mathcal{C}) \subset \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$, thus is an internal left adjoint by [32, proposition 3.32].

□

Notation 3.1.4. Let $\Gamma : \mathcal{P}_{\mathcal{V}}(\text{Span}_n(\mathcal{C})) \rightarrow \mathcal{V}$ be the right adjoint of $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$.

Remark 3.1.5. • By construction, $\Gamma : \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \rightarrow \mathcal{V}$ is the enriched Hom functor $\underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))}(*, -)$ where $*$ is the unit of $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\mathcal{V}})$, at the level of underlying space of objects, it is the final object $* \in \mathcal{C}$.

- Moreover, since the enriched Yoneda embedding $\mathcal{Y}_{\mathcal{V}}$ is $\mathcal{V}$ -fully faithful, for $x \in \text{Span}_n^{\text{enr}}(\mathcal{C}) \hookrightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$, $\Gamma(x) = \underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))}(*, x) \cong \underline{\text{Hom}}_{\text{Span}_n^{\text{enr}}(\mathcal{C})}(*, x)$. Recall that $\underline{\text{Hom}}_{\text{Span}_n^{\text{enr}}(\mathcal{C})}(*, x) \cong \text{Span}_{n-1}(\mathcal{C}/_x)$, and is enhanced into an object in $(n-1) \text{ Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow (n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} = \mathcal{V}$ by the $\text{B}^{n-2} \text{Span}_1(\mathcal{C})$ -action on $\text{Span}_{n-1}(\mathcal{C}/_x)$, through the symmetric monoidal $(\infty, n-1)$ functor $\text{B}^{n-2} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_{n-1}(\mathcal{C}) \xrightarrow{(-) \times x} \text{Span}_{n-1}(\mathcal{C}/_x)$.

In conclusion:

Corollary 3.1.6. *There exists a chain of morphisms in $\text{CAlg}(\widehat{\text{Cat}^{\mathcal{V}}})$:*

$$\text{Span}_n^{\text{enr}}(\mathcal{C}) \xrightarrow{\mathcal{Y}_{\mathcal{V}}} \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

, where $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$ is equipped with a right adjoint Γ inside the non-full subcategory $\text{LMod}_{\mathcal{V}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}^{\mathcal{V}}}$, and $\text{Span}_n^{\text{enr}}(\mathcal{C})$ lands in the full subcategory $n \text{ Cat}^{\text{Span}_n(\mathcal{C})} \hookrightarrow \text{Cat}^{\mathcal{V}}$.

3.2 A lax symmetric monoidal (∞, n) functor

In this subsection, we again denote the presentably symmetric monoidal category $(n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ as $\mathcal{V}$.

Construction 3.2.1. Consider the canonical symmetric adjunction

$$F : \mathbf{Spc} \rightleftarrows \mathcal{P}(\mathbf{Span}_1(\mathcal{C})) : G$$

where F is the unique symmetric monoidal colimit-preserving functor from $\mathbf{Spc}$ to $\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))$. Then there exists a symmetric monoidal adjunction

$$F_{n-1,*} : \mathbf{Cat}_{n-1} \rightleftarrows (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} : G_{n-1,*}$$

especially, $G_{n-1,*}$ can be enhanced into a lax symmetric monoidal functor.

Moreover, one can show that the above adjunction can be enhanced into an adjunction in $\widehat{\mathbf{Cat}}_n$: Since $F_* : \mathbf{Cat}_{n-1} \rightarrow (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$ is a symmetric monoidal colimit-preserving functor, it gives rise to a morphism $[1] \rightarrow \mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L))$. Consider the lax symmetric monoidal inclusion $\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_n$, which preserves adjunction by [6, corollary 4.19], thus we get a symmetric monoidal (∞, n) functor:

$$F_* : \mathbf{Cat}_{n-1} \rightarrow (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$$

in $\widehat{\mathbf{Cat}}_n$ and the $(\infty, 1)$ right adjoint $G_* : (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \rightarrow \mathbf{Cat}_{n-1}$ can be enhanced into a lax symmetric monoidal (∞, n) functor.

Here are some remarks one can make on the above construction:

Remark 3.2.2. • By the above construction, we enhance the presentably symmetric monoidal category $(n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$ into a symmetric monoidal (∞, n) category through the lax symmetric monoidal inclusion $\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_n$, especially, by forgetting the $\mathbf{Cat}_{n-1}$ -action, one recover the presentably symmetric monoidal $(\infty, 1)$ category structure on $(n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$

$$\begin{array}{ccc} \mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) & \longrightarrow & \widehat{\mathbf{Cat}}_n \\ \downarrow & & \downarrow \\ \mathbf{Pr}^L & \longrightarrow & \widehat{\mathbf{Cat}} \end{array}$$

.

- By replacing n with $n+1$, the forgetful functor

$$G_{n,*} : n \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \rightarrow \mathbf{Cat}_n$$

can be enhanced into a lax symmetric monoidal $\mathbf{Cat}_{n-1}$ -enriched functor (that is, a lax symmetric monoidal (∞, n) functor), thus preserving adjunction.

- By the corollary 5.3.6, there is a commutative diagram in $\mathbf{Op}$:

$$\begin{array}{ccc} \mathbf{LMod}_*(\mathbf{Cat}) & \xrightarrow{\cong} & \mathbf{Cat}^{\mathcal{P}(\ast)} \cong \mathbf{Cat} \\ \text{forget} \uparrow & & \uparrow G_* \\ \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}) & \longrightarrow & \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \end{array}$$

, thus for $\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{(n-1)\text{Cat}^\mathcal{V}})$, it recovers the symmetric monoidal (∞, n) -category $\text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$ under the transfer of enrichment along $G_{n-1,*} : (n-1)\text{Cat}^\mathcal{V} \rightarrow \text{Cat}_{n-1}$.

Now consider the following morphisms [corollary 3.1.6](#) in $\text{CAlg}(\text{Cat}^\mathcal{V})$, where $\mathcal{V} = (n-1)\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$

$$\text{Span}_n^{enr}(\mathcal{C}) \xrightarrow{\mathfrak{L}_\mathcal{V}} \mathcal{P}_\mathcal{V}(\text{Span}_n^{enr}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

Under the lax symmetric monoidal (∞, n) functor $G_{n,*} : \text{CAlg}(n\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}) = \text{CAlg}(\text{Cat}^\mathcal{V}) \rightarrow \text{CAlg}(\text{Cat}_n)$, there are morphisms:

$$\text{Span}_n(\mathcal{C}) \xrightarrow{\mathfrak{L}_\mathcal{V}} \mathcal{P}_\mathcal{V}(\text{Span}_n^{enr}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

where $\mathfrak{L}_\mathcal{V}$ is a symmetric monoidal (∞, n) functor and $1 : \mathcal{V} \rightarrow \mathcal{P}_\mathcal{V}(\text{Span}_n^{enr}(\mathcal{C}))$ admits the right adjoint Γ in Cat_n since $G_{n,*}$ preserves adjunction.

Remark 3.2.3. A priori, $\Gamma : \mathcal{P}_\mathcal{V}(\text{Span}_n^{enr}(\mathcal{C})) \rightarrow \mathcal{V}$ is a left $\mathcal{V}$ -module functor in $\text{LMod}_\mathcal{V}(\text{Pr}^L)$, thus is automatically a $\mathcal{V}$ -enriched functor. On the other hand, by the forgetful functor $G_{n-1,*} : \mathcal{V} \rightarrow \text{Cat}_{n-1}$, Γ can be regarded as an (∞, n) functor.

Thus Γ can be enhanced into a lax symmetric monoidal (∞, n) functor since 1 is a symmetric monoidal (∞, n) functor. And there is a chain of lax symmetric monoidal (∞, n) functors:

$$\text{Span}_n(\mathcal{C}) \rightarrow \mathcal{P}_\mathcal{V}(\text{Span}_n^{enr}(\mathcal{C})) \xrightarrow{\Gamma} \mathcal{V} = (n-1)\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

Remark 3.2.4. One can see that: for $\Gamma(x) = \underline{\text{Hom}}_{\text{Span}_n^{enr}(\mathcal{C})}(*, x) \in n\text{Cat}^{\text{Span}_1(\mathcal{C})} \cong (n-1)\text{Cat}^{\text{Cat}^{\text{Span}_1(\mathcal{C})}}$, it can be regarded as the image of $\text{Span}_{n-1}^{enr}(\mathcal{C}_{/x}) \in (n-1)\text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})}$ (by [corollary 2.6.7](#)) under the transfer of enrichment along the forgetful functor of slice category $\text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C})$:

- Recall the diagram $\square$ in the proof of [lemma 2.6.4](#). Combined with the commutative diagram in [corollary 5.3.6](#), we have:

$$\begin{array}{ccccc} \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x})}(\text{Cat})_{\text{closed}} & \hookrightarrow & \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x})}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}_{/x}))} \\ \downarrow & & \downarrow & & \downarrow g_* \\ \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} & \hookrightarrow & \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \end{array}$$

, where the middle vertical map is the forgetful functor along $(-) \times x : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x})$, and g_* is the right adjoint of the transfer of enrichment along $(-) \times t : \mathcal{P}(\text{Span}_1(\mathcal{C})) \rightarrow \mathcal{P}(\text{Span}_1(\mathcal{C}_{/x}))$. Noticing the biadjunction:

$$(-) \times t : \text{Span}_1(\mathcal{C}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : \text{forget along } x \rightarrow *$$

, taking the presheaf category gives rise to the adjunction

$$(-) \times t : \mathcal{P}(\text{Span}_1(\mathcal{C})) \rightleftarrows \mathcal{P}(\text{Span}_1(\mathcal{C}_{/x})) : \text{forget along } x \rightarrow *$$

. Thus, g_* is the transfer of enrichment along the forgetful functor $\mathcal{P}(\text{Span}_1(\mathcal{C}_{/x})) \rightarrow \mathcal{P}(\text{Span}_1(\mathcal{C}))$, and it restricts to $\text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$.

Notation 3.2.5. By the above remark, without abuse of notation, we again denote the image of $\text{Span}_{n-1}^{\text{enr}}(\mathcal{C}_{/x}) \in (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})}$ under the transfer of enrichment along forgetful functor $\text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C})$ again as $\text{Span}_n^{\text{enr}}(\mathcal{C}_{/x})$

Corollary 3.2.6. *There exists a lax symmetric monoidal (∞, n) functor:*

$$\text{Span}_n(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

assigning $x \in \text{Span}_n(\mathcal{C})$ with $\text{Span}_{n-1}^{\text{enr}}(\mathcal{C}_{/x})$ in [notation 3.2.5](#).

4 2 Shv(BG)

In this section, we fix $\mathcal{C} = \text{Stk}$, which is the category of Artin stacks of finite type with affine stabilizer over $\overline{k}_q$ where $q = p^n$ for a prime number p , and G is a smooth algebraic group over $\overline{k}_q$.

4.1 2 Shv(-)

Given a sheaf theory, which is a lax symmetric monoidal functor $\text{Shv}(-) : \text{Span}_1(\text{Stk}) \rightarrow 1 \text{ Pr}$,

- There is an induced colimit-preserving lax symmetric monoidal functor

$$\mathcal{P}(\text{Span}_1(\mathcal{C})) \rightarrow 1 \text{ Pr}$$

- There is an induced symmetric monoidal functor

$$\text{Env}(\text{Span}_1(\mathcal{C})) \rightarrow 1 \text{ Pr}$$

- There exists a localization:

$$l : \text{Env}(\mathcal{C}) \rightleftarrows \mathcal{C} : i$$

, which induces a localization:

$$\mathcal{P}(l) : \mathcal{P}(\text{Env}(\mathcal{C})) \rightleftarrows \mathcal{P}(\mathcal{C}) : \mathcal{P}(i)$$

, which is a symmetric monoidal adjunction, thus there is an induced symmetric monoidal adjunction by transfer of enrichment:

$$\mathcal{P}(l)_* : \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightleftarrows \text{Cat}^{\mathcal{P}(\mathcal{C})} : \mathcal{P}(i)_*$$

.

- Moreover, we can enhance the symmetric monoidal left adjoint $\mathcal{P}(l)_* : \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ into a symmetric monoidal $(\infty, 2)$ functor: The composition of morphisms $* \rightarrow \text{Env}(\mathcal{C}) \rightarrow \mathcal{C}$ in $\text{CAlg}(\text{Cat})$ induces a composition of morphisms $\mathcal{P}(*) \cong \text{Spc} \rightarrow \mathcal{P}(\text{Env}(\mathcal{C})) \rightarrow \mathcal{P}(\mathcal{C})$ in $\text{CAlg}(\text{Pr}^L)$; thus, there is an induced composition of morphisms $\text{Cat}^{\text{Spc}} \cong \text{Cat} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ in $\text{CAlg}(\text{Pr}^L)$; thus, $\text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ is a morphism in $\text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L)) \rightarrow \text{CAlg}(\text{Cat}_2)$.

- Thus, the right adjoint $\mathcal{P}(i)_* : \text{Cat}^{\mathcal{P}(\mathcal{C})} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))}$ can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.
- Likewise, the composition of morphisms $* \rightarrow \text{Env}(\mathcal{C}) \rightarrow 1 \text{ Pr}$ in $\text{CAlg}(\text{Cat})$ induces $\text{Spc} \rightarrow \mathcal{P}(\text{Env}(\mathcal{C})) \rightarrow 1 \text{ Pr}$ in $\text{CAlg}(\text{Pr}^L)$, and thus $\text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}(1 \text{ Pr})$ is a morphism in $\text{CAlg}(\text{Cat}_2)$.
- Thus there is a lax symmetric monoidal $(\infty, 2)$ functor $\text{Cat}^{\text{Span}_1(\mathcal{C})} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{1 \text{ Pr}}$.

And by composition with the lax symmetric monoidal (∞, n) functor constructed in [corollary 3.2.6](#):

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

, there is a lax symmetric monoidal $(\infty, 2)$ functor

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{1 \text{ Pr}}$$

.

Definition 4.1.1. Let $2 \text{ Shv}(-)$ denote the lax symmetric monoidal $(\infty, 2)$ functor:

$$2 \text{ Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{1 \text{ Pr}} \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}^L) \cong 2 \text{ Pr}$$

where $\mathcal{P}_{1 \text{ Pr}}(-)$ is the enriched presheaf functor.

Remark 4.1.2. Noticing the following commutative diagram:

$$\begin{array}{ccccc} \text{Cat}^{1 \text{ Pr}} & \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} & \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}) & \longrightarrow & \text{LMod}_{1 \text{ Pr}}(\text{Pr}^L) \\ (\text{Spc} \rightarrow 1 \text{ Pr})_* \uparrow & & \uparrow (-) \otimes 1 \text{ Pr} & & \uparrow (-) \otimes 1 \text{ Pr} \\ \text{Cat}^{\text{Spc}} \cong \text{Cat} & \xrightarrow{\mathcal{P}(-)} & 1 \text{ Pr} & \longrightarrow & \text{Pr}^L \end{array}$$

Especially, we can extract the following diagram in $\text{CAlg}(\text{Pr}^L)$:

$$\begin{array}{ccc} \text{Cat}^{1 \text{ Pr}} & \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} & \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}) \\ (\text{Spc} \rightarrow 1 \text{ Pr})_* \uparrow & & \uparrow (-) \otimes 1 \text{ Pr} \\ \text{Cat}^{\text{Spc}} \cong \text{Cat} & \xrightarrow{\mathcal{P}(-)} & 1 \text{ Pr} \end{array}$$

where $\mathcal{P}_{1 \text{ Pr}}$ preserves colimits by [32, proposition 6.40]. Thus $\mathcal{P}_{1 \text{ Pr}}(-) : \text{Cat}^{1 \text{ Pr}} \rightarrow \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr})$ can be enhanced into a symmetric monoidal $(\infty, 2)$ functor.

Remark 4.1.3. As a supplementary remark, for $\mathcal{V} \in \text{Pr}^\kappa$, $\mathcal{P}_{\mathcal{V}}(-) : \text{Cat}^{\mathcal{V}} \rightarrow \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ factors through the non-full subcategory $\text{LMod}_{\mathcal{V}}(\text{Pr}_\kappa^L)$: by construction of the enriched presheaf category in [16], $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$ is κ -compactly generated by $v \otimes m$ where $v \in \mathcal{V}^\kappa$ and $m \in \mathcal{M}$, since there is functorial equivalence

$$\text{Map}_{\mathcal{P}_{\mathcal{V}}(\mathcal{M})}(v \otimes m, -) \cong \text{Map}_{\mathcal{V}}(v, \underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\mathcal{M})}(m, -))$$

and m is $\mathcal{V}$ -atomic in $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$. Moreover, the action map $\mathcal{V} \otimes \mathcal{P}_{\mathcal{V}}(\mathcal{M})$ preserves κ -compact objects, thus the enriched presheaf category of $\mathcal{M} \in \text{Cat}^{\mathcal{V}}$ lands in $\text{LMod}_{\mathcal{V}}(\text{Pr}_{\kappa}^L)$. By the description of κ -compact generators in $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$, for any $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\text{Cat}^{\mathcal{V}}$, the induced morphism $\mathcal{P}_{\mathcal{V}}(\mathcal{M}) \rightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{N})$ in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ lands in $\text{LMod}_{\mathcal{V}}(\text{Pr}_{\kappa}^L)$.

Remark 4.1.4. Moreover, the $(\infty, 2)$ structure on 2Pr can be described as follows: given $\mathcal{M}, \mathcal{N} \in \text{LMod}_{1\text{Pr}}(1\text{Pr})$, the Hom category $\underline{\text{Hom}}_{2\text{Pr}}(\mathcal{M}, \mathcal{N}) \cong \text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1})$, that is, the category of $\aleph_1$ -continuous $(1\text{Pr})^{\aleph_1}$ -linear functors between $\mathcal{M}^{\aleph_1}$ and $\mathcal{N}^{\aleph_1}$.

Remark 4.1.5. Noticing the following symmetric monoidal adjunction of (large version) adjoining the $\aleph_1$ -filtered colimits:

$$\widehat{\text{Cat}} \rightleftarrows \widehat{\text{Cat}}^{\aleph_1}$$

where $\widehat{\text{Cat}}^{\aleph_1}$ is the very large category of large categories with small $\aleph_1$ -filtered colimits, and functors preserving small $\aleph_1$ -filtered colimits. Especially, the left adjoint restricts to the symmetric monoidal functor $\text{Ind}_{\aleph_1}(-) : \text{Cat} \rightarrow \widehat{\text{Cat}}^{\aleph_1}$; then there exists an induced symmetric monoidal adjunction:

$$\widehat{\text{Cat}}^{\widehat{\text{Cat}}} \rightleftarrows \widehat{\text{Cat}}^{\widehat{\text{Cat}}^{\aleph_1}}$$

, and the unit map valued at $2\text{Pr} \in \widehat{\text{Cat}}_2$ gives a map

$$2\text{Pr} \rightarrow 2\text{Pr}^{\#}$$

in $\text{CAlg}(\widehat{\text{Cat}}^{\widehat{\text{Cat}}}) \cong \widehat{\text{Cat}}_2$, where it changes the Hom category $\text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1})$ between $\mathcal{M}, \mathcal{N}$ in 2Pr to $\text{Ind}_{\aleph_1}(\text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1}))$, which is the internal Hom of the Cartesian closed symmetric monoidal 2Pr .

Remark 4.1.6. 4.2 $2\text{Shv}(\text{BG})$

In this subsection, we fix our sheaf theory as the one developed in [22], which is a lax symmetric monoidal functor:

$$\text{Shv}(-) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Pr}^L$$

assigning a stack $X \in \text{Stk}$ with the ∞ -category of l -adic sheaves $\text{Shv}(X)$ on X , and given a span

$$x \xleftarrow{f} z \xrightarrow{g} y \quad \text{it sends } f \text{ to } f^* : \text{Shv}(z) \rightarrow \text{Shv}(x), \text{ and } g \text{ to } g_! : \text{Shv}(z) \rightarrow \text{Shv}(y).$$

Remark 4.2.1. The l -adic sheaf theory we take $\text{Shv}(-) : \text{Span}(\text{Stk}) \rightarrow \text{Pr}^L$ lands inside the non-full subcategory 1Pr :

- For $\mathcal{X} \in \text{Stk}$ with atlas of a scheme $j : X \rightarrow \mathcal{X}$, $\text{Shv}(\mathcal{X})$ is compactly generated by objects in the form of $j_!(\mathcal{F})$ for compact object $\mathcal{F} \in \text{Shv}(X)^c$.
- For a morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in Stk , $f_!$ preserves compactness since $f^!$ is continuous.
- However, f^* doesn't preserve compactness in general (for example, $\frac{1}{\text{BG}_n}$ is not compact). Instead, f^* preserves constructibility (see its definition in [20, definition 2.1.9]), and a constructible sheaf on any $\mathcal{X} \in \mathcal{C}$ is $\aleph_1$ -compact. Thus, for an atlas $h : Y \rightarrow \mathcal{Y}$ and

compact $\mathcal{F} \in \mathrm{Shv}(Y)^c$, we take the $\aleph_1$ -compact generator $h_! \mathcal{F}$ of $\mathcal{Y}$, and by base change, we have $f^* h_! (\mathcal{F}) \cong h'_! (f')^* (\mathcal{F})$

$$\begin{array}{ccc} Y \times_{\mathcal{Y}} \mathcal{X} & \xrightarrow{f'} & Y \\ h' \downarrow & & \downarrow h \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

. Moreover, $(f')^* (\mathcal{F})$ is constructible, thus $\aleph_1$ -compact, and so is $h'_! (f')^* (\mathcal{F})$.

Definition 4.2.2. Given a sheaf theory $\mathrm{Shv}(-)$ and a morphism $f : x \rightarrow y$ in $\mathcal{C}$, with Čech nerve $x_{\bullet} \rightarrow y$ we say that f satisfies Shv-descent if $\mathrm{Shv}(y)$ is the colimit of $\mathrm{Shv}(x_{\bullet})$ under the lower $!$ transition map. If the pullback of f satisfies Shv-descent, we say f satisfies universal Shv-descent.

Example 4.2.3. Consider the étale sheaf theory $\mathrm{Shv}(-)$ developed in [22] and a morphism $f : * \rightarrow \mathrm{BG}$, then we have $\mathrm{Shv}(\mathrm{BG}) \cong \mathrm{Tot}(\mathrm{Shv}(*) \rightarrow \mathrm{Shv}(G) \cdots)$ can be computed as either upper $*$ or upper $!$, which differs by a shift. And the limit is equivalent to the colimit taken in 1 Pr by passing the right adjoint upper $!$ to left adjoint lower $!$. Thus $f : * \rightarrow \mathrm{BG}$ satisfies Shv-descent. Likewise, the above argument works for any pullback of $* \rightarrow \mathrm{BG}$, thus $f : * \rightarrow \mathrm{BG}$ satisfies universal Shv-descent.

Proposition 4.2.4. Given a morphism $f : x \rightarrow y \in \mathcal{C}$, if f satisfies universal Shv-descent, then $2 \mathrm{Shv}(y) \cong \mathrm{Tot}(2 \mathrm{Shv}(x_{\bullet}))$.

Proof. This is [37, proposition 4.2]. □

Thus we have the following diagram in $\mathrm{LMod}_{1 \mathrm{Pr}}(\mathrm{Pr}^L)$.

$$2 \mathrm{Shv}(\mathrm{BG}) \xrightarrow{\cong} \mathrm{Tot}(2 \mathrm{Shv}(*) \longrightarrow 2 \mathrm{Shv}(G) \rightrightarrows 2 \mathrm{Shv}(G^2) \rightrightarrows \cdots)$$

Recall the following proposition on descent and monadicity:

Proposition 4.2.5. Given a diagram $F : J \rightarrow \mathrm{Cat}_1$, assume that there exists an object $0 \in J$ such that for any object $x \in J$, there exists a morphism $0 \rightarrow x \in J$, and the structure morphism $s : \lim F \rightarrow F(0)$ admits a left adjoint L , then the adjunction

$$L : F(0) \rightleftarrows \lim F : s$$

is monadic.

Proof. This is [25, proposition 4.5.7] □

Corollary 4.2.6. The adjunction induced by the atlas map $f : * \rightarrow \mathrm{BG}$:

$$f^* : 2 \mathrm{Shv}(\mathrm{BG}) \rightleftarrows 2 \mathrm{Shv}(*) : f_*$$

is monadic, and the monad is the group object $G \in \mathcal{C}$. Thus there is an equivalence $2 \mathrm{Shv}(\mathrm{BG}) \cong \mathrm{LMod}_G(2 \mathrm{Shv}(*))$

Proof. • By construction, $2\text{Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow \text{LMod}_{1\text{Pr}}(\text{Pr}^L)$ gives us a biadjunction

$$f^* : 2\text{Shv}(\text{BG}) \rightleftarrows 2\text{Shv}(*) : f_*$$

where f^* sends an object $x \rightarrow \text{BG} \in 2\text{Shv}(\text{BG})$ to $x \times_{\text{BG}} *$ and f_* sends an object $y \rightarrow * \in 2\text{Shv}(*)$ to $y \rightarrow * \rightarrow \text{BG} \in 2\text{Shv}(\text{BG})$.

- By the above proposition, our adjunction is monadic.
- Moreover, consider the commutative diagram $* \rightarrow \text{BG} \rightarrow *$, which induces an equivalence:

$$\begin{array}{ccc} \text{BG} & \xleftarrow{f} & * \\ & \searrow s & \swarrow \\ & * & \end{array}$$

$$\begin{array}{ccc} 2\text{Shv}(\text{BG}) & \xrightarrow{f^*} & 2\text{Shv}(*) \\ & \searrow s^* & \swarrow \\ & 2\text{Shv}(*) & \end{array}$$

where $s : \text{BG} \rightarrow *$ is the structure map.

- The above commutative diagram makes $2\text{Shv}(\text{BG})$ as a $2\text{Shv}(*)$ -left module category, and the functor f^* is thus enhanced into a $2\text{Shv}(*)$ -left module functor. Thus the right adjoint f_* (noticing that there we have a biadjunction) is naturally enhanced into a lax $2\text{Shv}(*)$ -left module functor, which is in fact strict: since both f_* and f^* are colimit-preserving, thus it suffices to check the generators $y, x \rightarrow *$, which is obviously true.
- Thus, the monad is represented by an object $T \in \text{Alg}(2\text{Shv}(*)$, the underlying object of T is just $f^*f_*(*) \cong G \in 2\text{Shv}(*)$. And the algebra object structure on T is given by $T \times T \cong f^*(f_*f^*)f_*(*) \rightarrow f^*\text{id}f_*(*) \cong T$, which is the multiplication map for the group scheme G .

□

5 Appendix on enriched category theory

5.1 Enriched category theory

In this subsection, we review two different models of enriched category theory; one is categorical algebra by [10], and the other is a weakly left tensored category by [15]. These two models are equivalent, as indicated by [15].

Notation 5.1.1. Let $\text{Ass} := \Delta^{op}$ be the category of finite non-empty totally ordered sets and order preserving maps. We call a map $[n] \rightarrow [m] \in \text{Ass}$

- inert if it corresponds to a map of Δ in the form $[m] \cong \{i, i+1, \dots, i+m\} \subset [n]$ for some $i \geq 0$.
- standard inert if it is inert and $m = 1$.

- active if it corresponds to a map of Δ that preserves the minimum and maximum.

The following definitions follow [10, subsection 4] and [15, subsection 4.1].

Definition 5.1.2. Given a space $X \in \mathbf{Spc}$, let $\text{Ass}_X \rightarrow \text{Ass}$ be the coCartesian fibration classifying the functor $i_* X \in \text{Fun}(\text{Ass}, \text{Cat})$ where $i_* : \text{Cat} \rightarrow \text{Fun}(\text{Ass}, \text{Cat})$ is the right Kan extension along $[0] : * \rightarrow \text{Ass}$.

Definition 5.1.3. A categorical algebra in a non-symmetric operad $\mathcal{V}^\otimes \rightarrow \text{Ass}$ with underlying space of objects X is a morphism $\text{Ass}_X^{\text{op}} \rightarrow \mathcal{V}^\otimes$ over Ass preserving the coCartesian liftings over the inert maps in Ass .

One can show that $\text{Ass}_X \rightarrow \text{Ass}$ is a generalized non-symmetric operad defined in [10, 2.4]. Moreover, the underlying category of Ass_X is $(\text{Ass}_X)_{[1]} \cong X \times X$, and given a categorical algebra $\mathcal{C} : \text{Ass}_X \rightarrow \mathcal{V}^\otimes$, the induced morphism over $[1] \in \text{Ass}$ is a functor $\mathcal{C}(-, -) : X \times X \rightarrow \mathcal{V}$, which can be regarded as the enriched Hom of $\mathcal{C}$, assigning for any objects $x, y \in X$ an object $\mathcal{C}(x, y) \in \mathcal{V}$.

Thus, categorical algebra can be regarded as a prototype of an enriched category, and given a categorical algebra $\mathcal{C} : \text{Ass}_X \rightarrow \mathcal{V}^\otimes$, we call an object x in the space X an object in $\mathcal{C}$, also denoted as $x \in \mathcal{C}$.

And one can assemble $\text{Alg}_{\text{Ass}_X}(\mathcal{V})$ for all $X \in \mathbf{Spc}$ and all $\mathcal{V} \in$ together as a fibration over $\mathbf{Spc} \times \text{Op}^{ns}$:

Theorem 5.1.4. *There exists a map $\text{PreCat} \rightarrow \text{Op}^{ns} \times \mathbf{Spc}$, which is coCartesian fibration over Op^{ns} and Cartesian fibration over $\mathbf{Spc}$, and the fiber over $(\mathcal{V}^\otimes, X) \in \text{Op}^{ns} \times \mathbf{Spc}$ is the category $\text{Alg}_{\text{Ass}_X}(\mathcal{V})$ of categorical algebras in $\mathcal{V}$ with underlying space of objects X .*

Proof. This is [15, notation 4.40 and remark 4.41]. □

Notation 5.1.5. Given a morphism between ono-symmetric operad $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the coCartesian lifting of f in $\text{PreCat} \rightarrow \text{Op}^{ns}$ is called transfer of enrichment along f , and we denote it as:

$$f_*^{\text{cat}} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \text{Alg}_{\text{cat}}(\mathcal{W})$$

.

Notation 5.1.6. Given a non-symmetric operad $\mathcal{V}^\otimes \in \text{Op}^{ns}$, let $\text{Alg}_{\text{cat}}(\mathcal{V})$ denote the fiber product $\text{PreCat} \times_{\text{Op}^{ns}} \{\mathcal{V}^\otimes\}$, which is a Cartesian fibration over $\mathbf{Spc}$.

Definition 5.1.7. Given a monoidal category $\mathcal{V}^\otimes \in \text{Mon} \rightarrow \text{Op}^{ns}$, a morphism $\mathcal{C} \rightarrow \mathcal{D} \in \text{Alg}_{\text{cat}}(\mathcal{V})$ is $(\mathcal{V})$ -fully faithful if it's the Cartesian lifting of the Cartesian fibration $\text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \mathbf{Spc}$.

Now we introduce another model of enriched category by weakly left tensored categories:

Definition 5.1.8. Let $\mathcal{V}^\otimes \rightarrow \text{Ass}$ be an ∞ -operad and $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ a map of coCartesian fibrations relative to the inert maps of Ass that preserves the maximum.

We call $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ an ∞ -category weakly left tensored over $\mathcal{V}$ (or an ∞ -category with weak left $\mathcal{V}$ -action) if the following conditions hold:

- (1) for every $n \geq 0$ the map $[0] \cong \{n\} \subset [n]$ in Δ induces an equivalence

$$\theta : \mathcal{M}_{[n]}^\otimes \rightarrow \mathcal{V}_{[n]}^\otimes \times \mathcal{M}_{[0]}^\otimes$$

- (2) for every $X, Y \in \mathcal{M}^\otimes$ lying over $[m], [n] \in \text{Ass}$ the coCartesian lift $Y \rightarrow Y'$ of the map $[0] \cong \{n\} \subset [n]$ in Δ induces a pullback square:

$$\begin{array}{ccc} \mathcal{M}^\otimes(X, Y) & \longrightarrow & \mathcal{M}^\otimes(X, Y') \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes(\phi(X), \phi(Y)) & \longrightarrow & \mathcal{V}^\otimes(\phi(X), \phi(Y')) \end{array}$$

Definition 5.1.9. Let $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes, \phi' : \mathcal{M}'^\otimes \rightarrow \mathcal{V}'^\otimes$ be ∞ -categories with weak left action. We call a commutative diagram square of ∞ -categories over Ass

$$\begin{array}{ccc} \mathcal{M}^\otimes & \xrightarrow{\gamma} & \mathcal{M}'^\otimes \\ \phi \downarrow & & \downarrow \phi' \\ \mathcal{V}^\otimes & \xrightarrow{\rho} & \mathcal{V}'^\otimes \end{array}$$

a lax linear functor between weakly left tensored ∞ -categories if γ preserves coCartesian lifts of inert morphisms of Ass that preserves the maximum.

Notation 5.1.10. Let $\text{Cat}_{\infty/\text{Ass}}^{\max} \subset \text{Cat}_{\infty/\text{Ass}}$ be the subcategory of coCartesian fibrations relative to the collection of inert morphisms which preserves the maximum. Let

$$\omega\text{LMod} \subset \text{Op}^{ns} \times_{\text{Fun}(\{1\}, \text{Cat}_{\infty/\text{Ass}}^{\max})} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\max})$$

be the full subcategories of ∞ -categories with a weak left action over some ∞ -operad.

Remark 5.1.11. By [15, proposition 3.33] and [19, proposition 2.72], there is an equivalence $_{/\text{LM}^\otimes} \cong \omega\text{LMod}$. This implies that the definition of weakly left tensored categories agrees with the one [25, definition 4.2.1.12]. Moreover, under the identification of ωLMod with $\text{Op}_{/\text{LM}^\otimes}$, there is a non-full subcategory inclusion $\text{LMod} \rightarrow \omega\text{LMod}$, which corresponds to weakly left tensored categories $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ which is a morphism between coCartesian fibrations over Ass , and lax linear functors (γ, ρ) which preserve all coCartesian liftings.

Theorem 5.1.12. [15, 3.32] *The canonical functor $\omega\text{LMod} \rightarrow \text{Op}^{ns}$ is a biCartesian fibration.*

Definition 5.1.13. Let $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ be a weakly left tensored ∞ -category.

- (1) We say that ϕ exhibits $\mathcal{M}$ as (left) pseudo-enriched in $\mathcal{V}$ if $\mathcal{V}^\otimes \rightarrow \text{Ass}$ is a monoidal ∞ -category and for every $X, Y \in \mathcal{M}$ and $V_1, \dots, V_n \in \mathcal{V}$, the map

$$\text{Mul}_{\mathcal{M}}(V_1 \otimes \dots \otimes V_n, X; Y) \rightarrow \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y)$$

induced by the active morphism $(V_1, \dots, V_n) \rightarrow V_1 \otimes \dots \otimes V_n$ is an equivalence.

- (2) a morphism object of $X, Y \in \mathcal{M}$ is an object $\text{Mor}_{\mathcal{M}}(X, Y) \in \mathcal{V}$ together with a multimorphism $\alpha \in \text{Mul}_{\mathcal{M}}(\text{Mor}_{\mathcal{M}}(X, Y), X; Y)$ that induces for all objects $Z_1, \dots, Z_n \in \mathcal{V}$ an equivalence of spaces:

$$\text{Mul}_{\mathcal{V}}(Z_1, \dots, Z_n; \text{Mor}_{\mathcal{M}}(X, Y)) \cong \text{Mul}_{\mathcal{M}}(Z_1, \dots, Z_n, X; Y)$$

is an equivalence.

- (3) We say that $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ enriched in $\mathcal{V}$ if for every $X, Y \in \mathcal{M}$, there exists a morphism object $\text{Mor}_{\mathcal{M}}(X, Y)$.

Since we are mainly interested in the category enriched in a monoidal category, we restrict ourselves to the following notations:

Notation 5.1.14. (1) Let PLMod denote the non-full subcategory of ωLMod consisting of pseudo-

enriched categories and lax linear functors

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xrightarrow{f} & \mathcal{N}^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xrightarrow{\alpha} & \mathcal{W}^{\otimes} \end{array} \quad \text{where } \alpha \text{ is a monoidal functor.}$$

- (2) Let ωPLMod be the full subcategory of ωLMod consisting of pseudo-enriched categories. Noticing that $\text{PLMod} \rightarrow \text{Mon}$ is the pullback of ωPLMod along the inclusion $\text{Mon} \rightarrow \text{Mon}^{\text{lax}}$ (where Mon^{lax} is the full subcategory of Op^{ns} consisting of monoidal categories).
- (3) Let Enr denote the full subcategory of ωPLMod consisting of enriched categories.
- (4) Fix $\mathcal{V} \in \text{Op}^{\text{ns}}$, let $\text{Cat}^{\mathcal{V}}$ denote the fiber of $\text{Enr} \rightarrow \text{Op}^{\text{ns}}$ over $\mathcal{V}$.
- (5) Fix $\mathcal{V} \in \text{Mon}$, let $\text{PLMod}_{\mathcal{V}}$ denote the fiber of $\text{PLMod} \rightarrow \text{Mon}$ over $\mathcal{V}$.

And there is a comparison functor by [15, theorem 6.7, proposition 6.10]:

Theorem 5.1.15. *There exists a fully faithful embedding $i : \text{Enr} \hookrightarrow \text{PreCat}$ over Op^{ns} that admits a left adjoint loc .*

Moreover, one can describe the essential image of i by the ‘complete categorical algebra’:

Definition 5.1.16. [10, proposition 5.42] Given a monoidal category $\mathcal{V}^{\otimes}$, a categorical algebra in $\mathcal{V}$ is called complete if it’s local with respect to the morphism $E^1 \rightarrow E^0$ in $\text{Alg}_{\text{cat}}(\mathcal{V})$, where E^1 is $\text{Ass}_{\{0,1\}} \rightarrow \text{Ass} \xrightarrow{1_{\mathcal{V}}} \mathcal{V}^{\otimes}$, and E^0 is $\text{Ass}_{\{0\}} \rightarrow \text{Ass} \xrightarrow{1_{\mathcal{V}}} \mathcal{V}^{\otimes}$, and the map $E^1 \rightarrow E^0$ is induced by $\{0, 1\} \rightarrow \{0\}$.

Theorem 5.1.17. *The essential image of $i : \text{Enr} \hookrightarrow \text{PreCat}$ over a monoidal category $\mathcal{V} \in \text{Mon} \rightarrow \text{Op}^{\text{ns}}$ is precisely the complete categorical algebras in $\mathcal{V}$.*

Definition 5.1.18. Given a monoidal category $\mathcal{V}$, and a morphism $f : \mathcal{C} \rightarrow \mathcal{D} \in \text{Alg}_{\text{cat}}(\mathcal{V})$, f is essential surjective if for every object $y \in \mathcal{D}$, there exists $x \in \mathcal{C}$ and a map $E^1 \rightarrow \mathcal{D}$ connecting the $f(x)$ and y in $\mathcal{D}$.

Remark 5.1.19. By [15, corollary 6.13] and [10, theorem 5.6.6], the fully faithful embedding $i_{\mathcal{V}} : \text{Cat}^{\mathcal{V}} \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{V})$ for a monoidal category $\mathcal{V}$ is a localization with respect to $\mathcal{V}$ -fully faithful and essential surjective morphisms.

5.2 Fibrations

For practical usage, we mainly consider the enriched category theory over $\text{Mon}^{lax} \hookrightarrow \text{Op}^{ns}$

Theorem 5.2.1. $\text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Mon}^{lax}$ is a coCartesian fibration, and the left adjoint $\text{loc} : \text{PreCat} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax}$ preserves coCartesian edges.

Proof. By [10, lemma 5.7.5], the coCartesian lifting over a lax monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$ in Mon^{lax} , which is the transfer of enrichment along f , preserves enriched fully faithful and essential surjective morphisms. Thus, by [25, lemma 2.2.4.11], $\text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Mon}^{lax}$ is a coCartesian fibration, and L preserves the coCartesian edges. $\square$

Notation 5.2.2. Given a morphism $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, we again call the coCartesian lifting over f the transfer of enrichment along f , denoted as

$$f_* : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{W}}$$

.By construction, this is the composition of $\text{Cat}^{\mathcal{V}} \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{V}) \xrightarrow{f_*^{\text{cat}}} \text{Alg}_{\text{cat}}(\mathcal{W}) \xrightarrow{\text{loc}_{\mathcal{W}}} \text{Cat}^{\mathcal{W}}$ where the arrow in the middle is the transfer of enrichment along f for the categorical algebras.

Proposition 5.2.3. Given a monoidal functor $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$ in Mon , if it admits a right adjoint $g : \mathcal{W}^\otimes \rightarrow \mathcal{V}^\otimes$ in Mon^{lax} , then there exists an adjunction

$$f_* : \text{Cat}^{\mathcal{V}} \rightleftarrows \text{Cat}^{\mathcal{W}} : g_*$$

Proof. For the model of weakly left tensored categories, this is by [16, 4.70]. For the model of categorical algebras, this is by [10, 5.7.17]. $\square$

Proposition 5.2.4. Consider the Cartesian fibration ωLMod over Op , it restricts to a Cartesian fibration PLMod over Mon , and the inclusion $\text{PLMod} \rightarrow \omega\text{LMod}$ preserves the Cartesian edges.

Proof. It suffices to show the Cartesian edge in ωLMod at a point $N^\otimes \rightarrow B^\otimes \in \text{PLMod}$ remains inside PLMod .

By [16, 2.62], the Cartesian edge in ωLMod over a map of operads α is just the pullback of operads: Given two weakly left tensored categories $M^\otimes \rightarrow A^\otimes$ and $N^\otimes \rightarrow B^\otimes$, an morphism of weakly left tensored modules $\alpha^* : M^\otimes \rightarrow N^\otimes$ is Cartesian iff the following diagram is a pullback diagram:

$$\begin{array}{ccc} M^\otimes & \xrightarrow{\alpha^*} & N^\otimes \\ \downarrow & \lrcorner & \downarrow \\ A^\otimes & \xrightarrow{\alpha} & B^\otimes \end{array}$$

, that is, for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, the following map induced is an equivalence:

$$\text{Mul}_M(a_1, \dots, a_n, m; n) \rightarrow \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n))$$

Thus if $\alpha : A \rightarrow B$ is a monoidal functor, then if N is pseudo-enriched in A , and for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, we have the following commutative diagram:

$$\begin{array}{ccc}
\text{Mul}_M(a_1 \otimes \cdots \otimes a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1 \otimes \cdots \otimes a_n), \alpha^*(m); \alpha^*(n)) \\
\downarrow & & \downarrow \\
\text{Mul}_M(a_1, \dots, a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n))
\end{array}$$

where the vertical arrows are induced by the active map corresponding to the tensor product. Thus the left vertical map is equivalence, and this implies M is pseudo-enriched in A . $\square$

Remark 5.2.5. The above statement is not true if the morphism $\alpha : A \rightarrow B$ is in Op instead of Mon , for example, by [16, corollary 4.22], the pullback along the lax monoidal functor $\mathcal{V} \rightarrow \text{Env}(\mathcal{V})$ takes the categories pseudo-enriched in $\text{Env}(\mathcal{V})$ to weakly left tensored categories over $\mathcal{V}$.

Proposition 5.2.6. *The (non-full) subcategory inclusion $\text{LMod} \rightarrow \text{PLMod}$ preserves Cartesian edges over Mon .*

Proof. We have already known $\text{LMod} \rightarrow \text{Mon}$ is a Cartesian fibration by [25, corollary 4.2.3.2], thus it suffices to show:

- Given a left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, and a monoidal functor $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the pullback in ωLMod :

$$\begin{array}{ccc}
\mathcal{M}^\otimes & \xrightarrow{f!} & \mathcal{N}^\otimes \\
\downarrow \lrcorner & & \downarrow \\
\mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes
\end{array}$$

lies in $\text{LMod} \subset \text{PLMod}$, which is true since pullback of coCartesian fibration is again coCartesian fibration.

- For any $\mathcal{N}' \in \text{LMod}_{\mathcal{V}}(\text{Cat})$, and a linear functor $\begin{array}{ccc} \mathcal{N}'^\otimes & \xrightarrow{F} & \mathcal{N}^\otimes \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes \end{array}$, there is a unique

$\mathcal{V}$ -linear functor $G : \mathcal{N}' \rightarrow \mathcal{M}$ making the following diagram

$$\begin{array}{ccccc}
& & \mathcal{N}'^\otimes & & \\
& \searrow & \downarrow & \searrow F & \\
& & \mathcal{M}^\otimes & \xrightarrow{f!} & \mathcal{N}^\otimes \\
& \searrow G & \downarrow & & \downarrow \\
& & \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes
\end{array}$$

, which is also true, since pullback in $\text{coC}/_{\text{Ass}}$ can be computed in $\text{Cat}/_{\text{Ass}}$, so there is a unique induced morphism in $\text{coC}/_{\text{Ass}} \mathcal{N}^{\otimes} \rightarrow \mathcal{M}^{\otimes}$ making the diagram commute, thus preserve coCartesian edges, and is a linear functor.

- Thus $f_!$ is locally Cartesian in $\text{LMod} \rightarrow \text{Mon}$, and thus Cartesian.

□

5.3 Symmetric monoidal structure

By [15, lemma 7.10], the coCartesian fibration $\text{PreCat} \rightarrow \text{Op}^{ns}$ preserves finite products; thus, there is an operad map:

$$\text{PreCat}^{\times} \rightarrow (\text{Op}^{ns})^{\times}$$

Moreover, the operad map is a coCartesian fibration by [15, remark 7.11].

Remark 5.3.1. Let $\boxtimes$ denote the product in PreCat , which, by [15, lemma 7.10], we can describe $F \boxtimes G$ for categorical algebras $F : \text{Ass}_X \rightarrow \mathcal{V}^{\otimes}$ and $G : \text{Ass}_Y \rightarrow \mathcal{W}^{\otimes}$ as:

$$\text{Ass}_{X \times Y} \cong \text{Ass}_X \times_{\text{Ass}} \text{Ass}_Y \rightarrow \mathcal{V}^{\otimes} \times_{\text{Ass}} \mathcal{W}^{\otimes}$$

, which is a categorical algebra in $\mathcal{V} \times \mathcal{W}^{\otimes}$.

Likewise, $\text{Enr} \rightarrow \text{Op}^{ns}$ also preserves finite products; thus, it can be enhanced into an operad map:

$$\text{Enr}^{\times} \rightarrow (\text{Op}^{ns})^{\times}$$

.

Given a symmetric monoidal category $\mathcal{V}$, it can be regarded as a commutative algebra object in Op^{ns} , specified by a map:

$$\{\mathcal{V}^{\otimes}\} : \text{Comm}^{\otimes} \rightarrow (\text{Op}^{ns})^{\times}$$

Notation 5.3.2. Given a symmetric monoidal category $\mathcal{V}$,

- (1) Let $\text{Alg}_{\text{cat}}(\mathcal{V})^{\otimes}$ denote the fiber product in Op :

$$\begin{array}{ccc} \text{Alg}_{\text{cat}}(\mathcal{V})^{\otimes} & \longrightarrow & \text{PreCat}^{\times} \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & (\text{Op}^{ns})^{\times} \end{array}$$

- (2) Let $\text{Enr}_{\mathcal{V}}^{\otimes}$ denote the fiber product in Op :

$$\begin{array}{ccc} (\text{Cat}^{\mathcal{V}})^{\otimes} & \longrightarrow & \text{Enr}^{\times} \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & (\text{Op}^{ns})^{\times} \end{array}$$

- (3) By [17, proposition 4.26], $\text{PLMod}^\times \rightarrow \text{Mon}^\times$ is also a coCartesian fibration, thus let $\text{PLMod}_\mathcal{V}^\otimes$ denote the fiber product in Op :

$$\begin{array}{ccc} \text{PLMod}_\mathcal{V}^\otimes & \longrightarrow & \text{PLMod}^\times \\ \downarrow & \lrcorner & \downarrow \\ \text{Comm}^\otimes & \xrightarrow{\{\mathcal{V}^\otimes\}} & \text{Mon}^\times \end{array}$$

Proposition 5.3.3. *Given a symmetric monoidal category $\mathcal{V}$, there is a symmetric monoidal adjunction:*

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes \rightleftarrows (\text{Cat}^\mathcal{V})^\otimes : i_\mathcal{V}$$

Moreover, given a symmetric monoidal adjunction

$$L : \mathcal{V}^\otimes \rightleftarrows \mathcal{W}^\otimes : R$$

Transfer of enrichment along L is again symmetric monoidal; thus, there is a symmetric monoidal adjunction:

$$L_*^{\text{cat}} : \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes \rightleftarrows \text{Alg}_{\text{cat}}(\mathcal{W})^\otimes : R_*^{\text{cat}}; L_* : \text{Cat}^\mathcal{V} \rightleftarrows \text{Cat}^\mathcal{W} : R_*$$

. And there is a commutative diagram:

$$\begin{array}{ccc} \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes & \xrightarrow{L_*^{\text{cat}}} & \text{Alg}_{\text{cat}}(\mathcal{W})^\otimes \\ \text{loc}_\mathcal{V} \downarrow & & \downarrow \text{loc}_\mathcal{W} \\ (\text{Cat}^\mathcal{V})^\otimes & \xrightarrow{L_*} & (\text{Cat}^\mathcal{W})^\otimes \end{array}$$

Proof. Consider the localization

$$\text{loc} : \text{PreCat}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times \rightleftarrows \text{Enr}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times : i$$

. First, we show that loc preserves all coCartesian liftings over $\text{Mon}^\times$: it suffices to show that it preserves the coCartesian liftings over the active map in $\text{Mon}^\times$: consider an active map $(\mathcal{V}, \mathcal{W}) \rightarrow \mathcal{V} \times \mathcal{W}^\otimes$, given localization maps of categorical algebra $\mathcal{M}$ in $\mathcal{V}$ and $\mathcal{N}$ in $\mathcal{W}$, it remains to show that the induced map $\mathcal{M} \boxtimes \mathcal{N} \rightarrow (L\mathcal{M} \boxtimes L\mathcal{N})$ is a local equivalence (fully faithful and essential surjective) in PreCat , which is true by the description of $\boxtimes$ in [remark 5.3.1](#).

Thus the induced map by taking pullback of $\text{PreCat}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times$ along $\{\mathcal{V}^\otimes\} : \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ is symmetric monoidal, and the adjunction between underlying categories can be enhanced into a symmetric monoidal adjunction:

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightleftarrows (\text{Cat}^\mathcal{V})^\otimes$$

Let $[1] \times \text{Comm}^\otimes \rightarrow (\text{Op}^{ns})^\times$ encode the symmetric monoidal functor L , then by pullback

along the coCartesian fibrations $loc : \text{PreCat}^\times$ and $\text{Enr}^\times$ over $(\text{Op}^{ns})^\times$, the acquired coCartesian fibration over $[1] \times \text{Comm}^\otimes$ classifies the symmetric monoidal functor L_*^{cat} and L_* . And the right adjoints R_*^{cat} and R_* can be enhanced into lax symmetric monoidal functors by [25, corollary 7.3.2.7].

□

Notation 5.3.4. (1) Let $\text{PPreCat}^\otimes$ denote the full suboperad of the fiber product $\widehat{\text{PreCat}}^\times \times_{\widehat{\text{Mon}}^\times} \text{Mon}^\times$ along $\mathcal{P}(-) : \text{Mon}^\times \rightarrow \widehat{\text{Mon}}^\times$, consisting of those categorical algebras with small underlying spaces of objects.

(2) Let $\text{PEnr}^\otimes$ denote the full suboperad of the fiber product $\widehat{\text{Enr}}^\times \times_{(\widehat{\text{Op}^{ns}})^\times} \text{Mon}^\times$ along $\mathcal{P}(-) : \text{Mon}^\times \rightarrow \widehat{\text{Mon}}^\times$, consisting of those enriched categories with small underlying spaces of objects.

By [15, corollary 6.13] and [15, lemma 7.14], there is:

Proposition 5.3.5. $\text{PPreCat}^\otimes \rightarrow \text{Comm}^\otimes$ is a Cartesian symmetric monoidal category, and moreover, there is a symmetric monoidal embedding $\text{PLMod}^\times \hookrightarrow \text{PPreCat}^\times$ over $\text{Mon}^\times$ which essential image consists of complete categorical algebras. Thus the symmetric monoidal full subcategory $\text{PEnr}^\otimes \hookrightarrow \text{PPreCat}^\otimes$ is also of Cartesian symmetric monoidal structure, and there is an induced symmetric monoidal equivalence $\text{PLMod}^\times \cong \text{PEnr}^\otimes$.

Now consider the lax symmetric monoidal inclusion $\text{LMod}^\times \rightarrow \text{PLMod}^\times$ over $\text{Mon}^\times$; we have the following corollary by [15, theorem 7.21]:

Corollary 5.3.6. Given a symmetric monoidal category $\mathcal{V}$, there is a lax symmetric monoidal inclusion $\text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes \rightarrow \text{PLMod}_{\mathcal{V}}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes$. Moreover, given a symmetric monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$, the above inclusion is contravariant:

$$\begin{array}{ccc} \text{LMod}_{\mathcal{V}}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\mathcal{V})} \\ \uparrow & & \uparrow \\ \text{LMod}_{\mathcal{W}}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\mathcal{W})} \end{array}$$

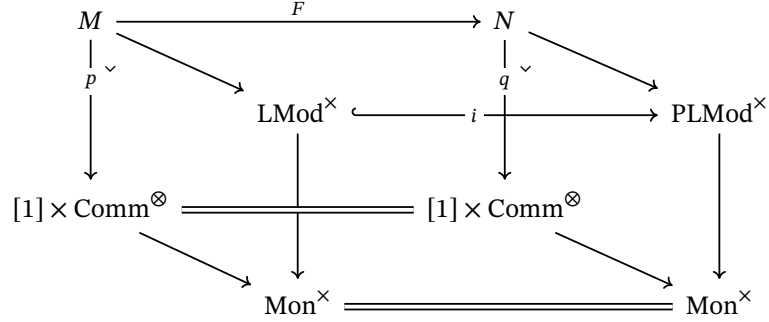
where the left vertical map is the forgetful functor along $\mathcal{V} \rightarrow \mathcal{W}$ and the right vertical map is the right adjoint of the transfer of enrichment along $\mathcal{P}(f) : \mathcal{P}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{W})$.

Proof. • Consider the symmetric monoidal inclusion $\text{LMod}^\times \rightarrow \text{PLMod}^\times \rightarrow \text{PPreCat}$ over $\text{Mon}^\times$. By taking the pullback along $\{\mathcal{V}^\otimes\} : \text{Comm}^\otimes \rightarrow \text{Mon}^\times$, there is a lax symmetric monoidal inclusion

$$\text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes \rightarrow \text{PLMod}_{\mathcal{V}}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes \hookrightarrow \text{PreCat}_{\mathcal{V}}^\otimes$$

- Since $\text{PLMod} \cong \text{PEnr}$ is Cartesian over Mon by [proposition 5.2.4](#), and also coCartesian since it's the pullback of the coCartesian fibration $\text{Enr} \rightarrow \text{Op}^{ns}$, thus $\text{PLMod} \rightarrow \text{Mon}$ is a biCartesian fibration.

- Let $\eta : [1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ encode the symmetric monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$, then consider the pullback along the morphism $\eta : [1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$. This gives rise to a morphism F between bifibrations over $[1] \times \text{Comm}^\otimes$ (Cartesian over $[1]$ and coCartesian over $\text{Comm}^\otimes$), as illustrated in the following diagram:



- The restriction of F to the fibers over $\{0\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the operad map $\text{LMod}_{\mathcal{V}}^\otimes \rightarrow \text{PLMod}_{\mathcal{V}}^\otimes$.
- The restriction of F to the fibers over $\{1\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the lax symmetric monoidal functor $R : \text{LMod}_{\mathcal{W}}(\text{Cat})^\otimes \hookrightarrow \text{PLMod}_{\mathcal{W}}^\otimes$.
- M and N are **biCartesian fibrations** over $[1]$ and **coCartesian fibrations** over $\text{Comm}^\otimes$ since pullback preserves Cartesian and coCartesian fibrations.
- $F : M \rightarrow N$ is a functor preserving Cartesian edges over $[1]$ and preserves the coCartesian edges over the inert maps in $\text{Comm}^\otimes$.
- Consider the fibration $M \rightarrow [1] \times \text{Comm}^\otimes$ which is biCartesian fibration over $[1]$ and coCartesian fibration over $\text{Comm}^\otimes$, by projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [14, 2.2.4] :

$$f_! : \text{LMod}_{\mathcal{V}}^\otimes \rightleftarrows \text{LMod}_{\mathcal{W}}(\text{Cat})^\otimes : f^!$$

, where $f_!$ is the symmetric monoidal ∞ -functor $(-) \otimes_{\mathcal{V}} \mathcal{W}$, and $f^!$ is the forgetful functor along $\mathcal{V} \rightarrow \mathcal{W}$.

- Consider the fibration $N \rightarrow [1] \times \text{Comm}^\otimes$, which is a biCartesian fibration over $[1]$ and a coCartesian fibration over $\text{Comm}^\otimes$. By projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [14, 2.2.4] :

$$f_* : \text{PLMod}_{\mathcal{V}}^\otimes \rightleftarrows \text{PLMod}_{\mathcal{W}}^\otimes : f^*$$

, where f_* is the transfer of enrichment along $\mathcal{P}(f)$ under the equivalence $\text{PLMod}_{\mathcal{V}}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes$ and $(\text{PLMod}_{\mathcal{W}})^\otimes$.

- Since F preserves the Cartesian edges over $[1]$, by [14, 2.2.4], the straightening gives us a morphism $\eta : [1] \rightarrow \text{Fun}([1]^{op}, \text{Cat}_{/\text{Comm}^\otimes})$, this is equivalent to a functor $[1] \times [1]^{op} \rightarrow \text{Cat}_{/\text{Comm}^\otimes}$, as illustrated in the following diagram in $\text{Cat}_{/\text{Comm}^\otimes}$:

$$\begin{array}{ccc}
\mathrm{LMod}_{\mathcal{W}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{W}}^{\otimes} \\
\downarrow f^! & & \downarrow f^* \\
\mathrm{LMod}_{\mathcal{V}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{V}}^{\otimes}
\end{array}$$

□

5.4 Loop and deloop

Consider the functor $\mathrm{PreCat} \rightarrow \mathrm{Op}^{ns} \times \mathrm{Spc}$ which is coCartesian fibration over Op^{ns} and Cartesian over Spc , by taking the fiber over $* \in \mathrm{Spc}$, by the construction of Quiv in [15, notation 4.37], $\mathrm{PreCat} \times_{\mathrm{Spc}} *$ is the coCartesian fibration over Op^{ns} classifying the functor:

$$\mathrm{Alg}_{\mathrm{Ass}}(-) : \mathrm{Op}^{ns} \rightarrow \mathrm{Cat}$$

which is the right adjoint of $(-) \times \mathrm{Ass} : \mathrm{Cat} \rightarrow \mathrm{Op}^{ns}$.

Notation 5.4.1. Let Alg denote the above fiber product $\mathrm{PreCat} \times_{\mathrm{Spc}} *$.

Moreover, one can consider the symmetric monoidal structure on the above construction: Given that the fiber embedding $\mathrm{PreCat}_0 \hookrightarrow \mathrm{PreCat}$ preserves products in PreCat by the description of $\boxtimes$ in remark 5.3.1, there is a symmetric monoidal functor

$$\mathrm{Alg}^{\times} \hookrightarrow \mathrm{PreCat}^{\times}$$

over $(\mathrm{Op}^{ns})^{\times}$.

Construction 5.4.2. Given a symmetric monoidal category $\mathcal{V}$, by pullback along $\{\mathcal{V}^{\otimes}\} : \mathrm{Comm}^{\otimes} \rightarrow (\mathrm{Op}^{ns})^{\times}$, there is a symmetric monoidal functor:

$$\mathrm{B} : \mathrm{Alg}(\mathcal{V})^{\otimes} \hookrightarrow \mathrm{PreCat}_{\mathcal{V}}^{\otimes} \xrightarrow{\mathrm{loc}_{\mathcal{V}}} \mathrm{Cat}^{\mathcal{V}}$$

where $\mathrm{Alg}(\mathcal{V})^{\otimes}$ is the pullback of $\mathrm{Alg}^{\times} \rightarrow (\mathrm{Op}^{ns})^{\times}$ along $\{\mathcal{V}\}$.

Remark 5.4.3. By [30, theorem 0.1], the symmetric monoidal category $\mathrm{Alg}(\mathcal{V})^{\otimes}$ in the above construction is equivalent to the commutative algebra object structure on the image of $\mathcal{V} \in \mathrm{CAlg}(\mathrm{Op}^{ns})$ in Cat under the lax symmetric monoidal functor $\mathrm{Alg}_{\mathrm{Ass}}(-) : \mathrm{Op}^{ns} \rightarrow \mathrm{Cat}$.

Theorem 5.4.4. [10, theorem 6.32] *Given a symmetric monoidal category $\mathcal{V}$, the symmetric monoidal functor B in construction 5.4.2 is fully faithful, and admits a right adjoint Ω .*

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